

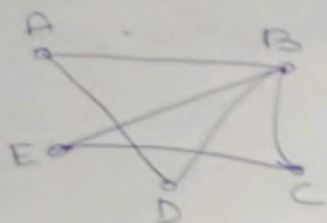
10/10/20

GRAPH THEORY

1) Represent the following scenario using a graph.

Let A, B, C, D and E be people. A knows B and D, B knows C and D, E, C knows E.

=>

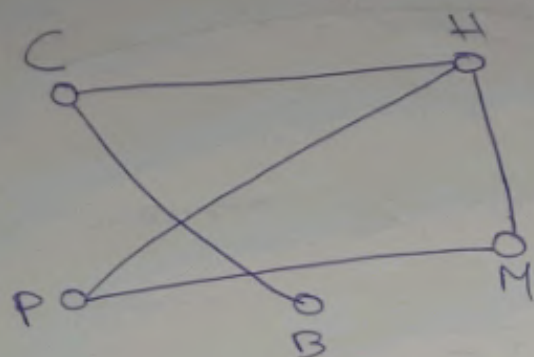


10/10/20

2) A country has a railway system between the following cities: Chennai, Bangalore, Hyderabad, Mumbai and Pune. There are 3 trains per day from Chennai to Bangalore, 2 trains from Bangalore to Chennai, 1 train from Chennai to Hyderabad, ~~1 train~~ 2 trains from Hyderabad to Mumbai, 1 train from Mumbai to Pune and 2 trains from Pune to Hyderabad.

3) Model this network using a graph where an edge exists between cities that have at least one train service in either direction.

⇒



$$3: C \rightarrow B$$

$$2: B \rightarrow C$$

$$1: C \rightarrow H$$

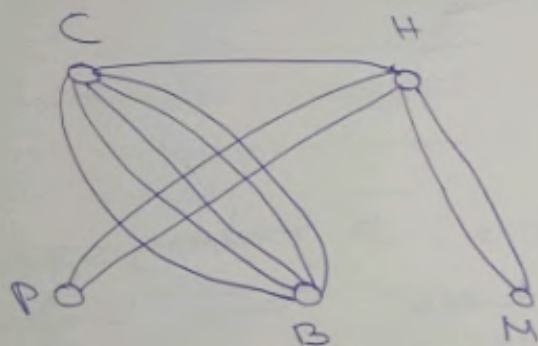
$$2: H \rightarrow M$$

$$1: M \rightarrow P$$

$$2: P \rightarrow H$$

ii) Model this network using a graph where each edge represents one train operating between cities (in either direction)

⇒

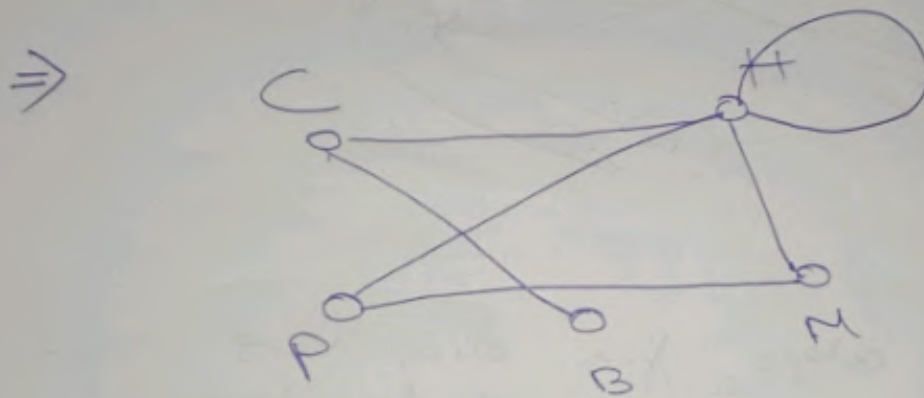


> ~~multiple~~ when there are ^{more than} one edge connecting two vertices the edges are called multiple edges / Parallel edges.

> graph that contain multiple edges are called multi graphs.

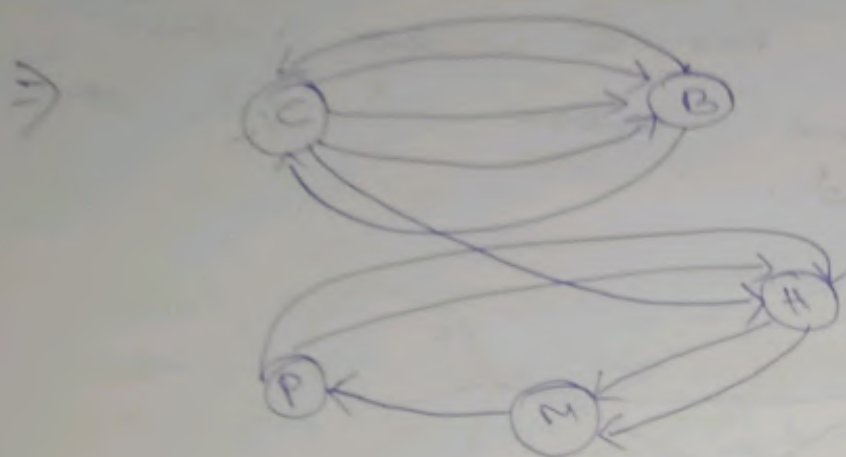
> graph without multiple edges or loops are called simple graph.

- iii) Model this network using a graph where an edge exists between cities that have at least one train service in either direction plus a loop for a circular tourist train that starts and ends at Hyderabad.



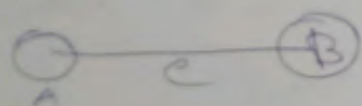
- > The edges that start and ends at the same vertex is known as a loop.
- > Any graph that contains loops are called Pseudo graph.
- > Pseudo graph may or maynot contain multiple edges.

ii) model his network using a graph where edges indicate the direction of the train.



> If the edges has direction then the graphs are called directed graphs. or they are called undirected graphs.

DEFINITION TERMS



end vertices

Edge e connects two vertices A and B then A and B are called end vertices of e . In such cases we also say e is incident on A and B . Since e is an

edge connecting A and B, A and B
will be called adjacent vertices

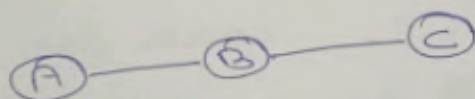
21/7/25
Monday

Informal Introduction to Graphs

A graph is a pictorial representation of relationships b/w objects. It helps us model connections, such as:

- > Friendships in social networks
- > routes between cities
- > Flight connections
- > network topologies in computers.

eg: consider 3 cities A, B and C. There are roads b/w A-B and B-C. we can represent this as



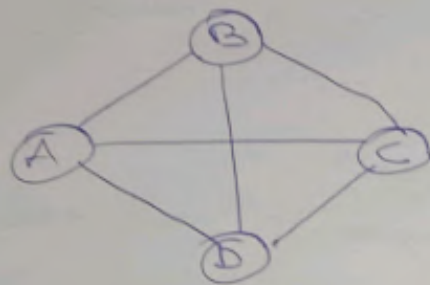
The Königsberg Bridge Problem.

In the 18th century, the city of Königsberg had seven bridges connecting different parts of land separated by a river. People wondered:

"Is it possible to take a walk that crosses each bridge exactly once and return to the starting point?"

Euler's insight: Represent the land areas as vertices and the bridges as edges, forming a graph. He showed that such a walk was not possible, laying the foundation of graph theory.

Graph Representation



Formal Definitions

Graph:

A graph G is an ordered pair $G = (V, E)$ where V is a non-empty set of vertices and E is a set of edges (unordered or ordered pairs from V).

Finite and Infinite Graph

A finite graph has a finite number of vertices and edges. An infinite graph has an infinite number

of vertices and edges.

Simple graph

An undirected graph with no loops and no multiple edges b/w the same pair of vertices.

Multigraph

An undirected graph that may have multiple edges between the same pair of vertices but no loops.

Pseudograph

A graph that may contain both multiple edges and loops.

Undirected Graph

A graph where the edges are unordered pair of vertices i.e., edge $\{u, v\} = \{v, u\}$.

Directed graph

A graph where the edges are ordered pair (arcs), i.e., (u, v) is directed from u to v .

Mixed graph

A graph that contains both ~~directed~~ directed and undirected edges.

example

1) Draw a graph that models the following

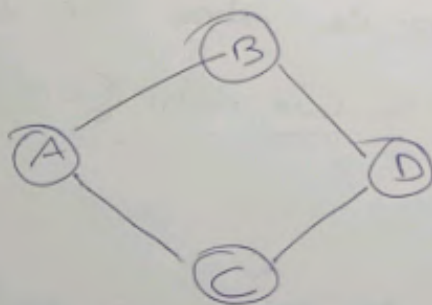
Situation : Let A, B, C, D be people.

A ~~is~~ Friends with B and C. B is also Friends with D. C is Friends with D.

Draw an undirected graph representing this.

Vertices : A, B, C, D.

Edges : $\{A, B\}, \{A, C\}, \{B, D\}, \{C, D\}$



BASIC TERMS IN GRAPH THEORY

- > **Loop**: A loop is an edge that connects a vertex to itself.
- > **Multiple edges**: Multiple edges are two or more edges that connect the same pair of vertices.
- > **Endpoints of an edge**: The endpoints of an edge are the two vertices that an edge connects.
- > **Adjacency**: Two vertices u and v in an undirected graph G are said to be adjacent if there is an edge $e = \{u, v\}$ in G that connects u and v .
- > **Incidence**: An edge e is said to be incident with a vertex v if v is one of the endpoints of e .
- > **Degree**: The degree of a vertex v , denoted $\deg(v)$, is the number of edges incident with v in an undirected graph.

> Isolated vertex : A vertex with degree zero (not connected to any other vertex) is called an isolated vertex.

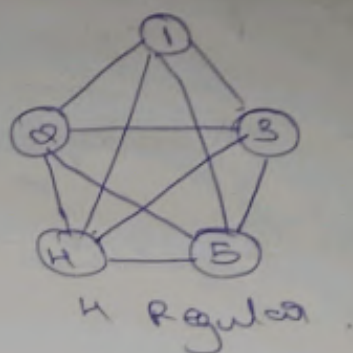
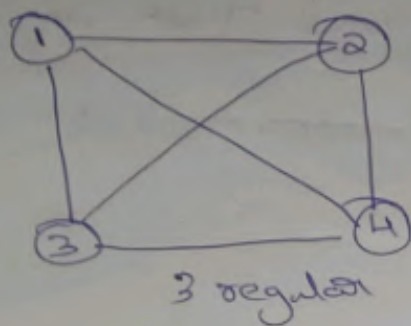
> Pendant vertex : A vertex with degree one is called a pendant vertex / Leaf.

> Initial and Terminal vertex (Directed graph) :

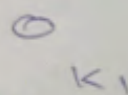
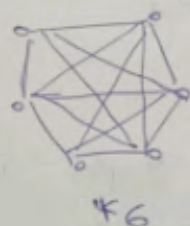
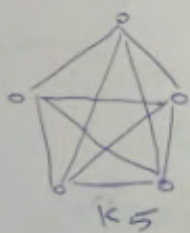
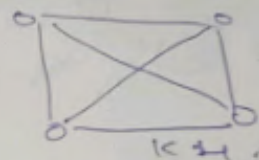
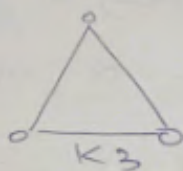
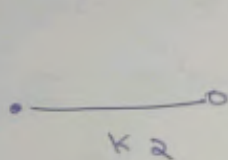
In a directed graph, for an edge (u, v) , u is the initial vertex and v is the terminal vertex.

> In-degree and out-degree : In a directed graph, the in-degree of a vertex is the number of edges coming into the vertex, and the out-degree is the number of edges going out of the vertex.

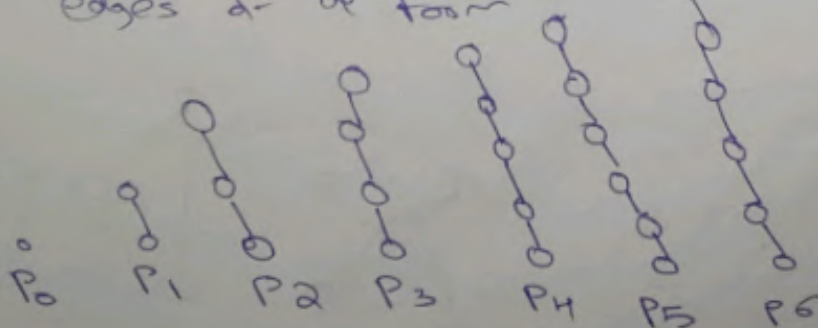
> Regular Graph : A regular graph is a simple graph in which every vertex has the same degree. If every vertex has degree x , then the graph is called an x -regular graph.



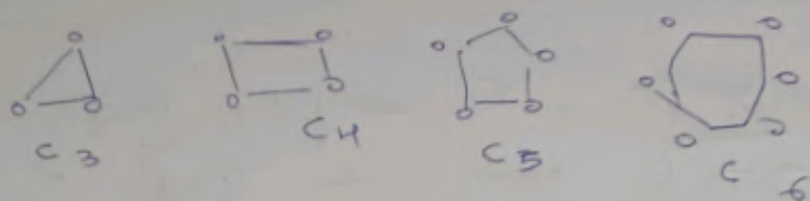
> Complete graph : A complete graph K_n , is a simple undirected graph in which every pair of distinct vertices is connected by a unique edge.



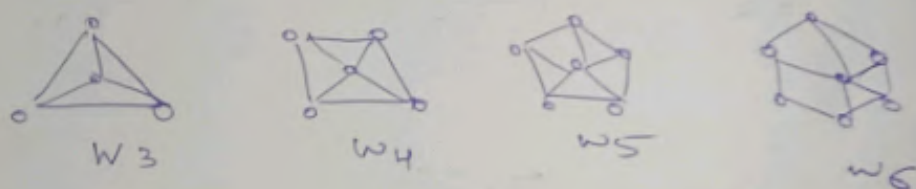
> ~~Path~~ P_n A Path P_n is a simple graph consisting of n vertices $v_1, v_2, \dots, v_n$ and $n-1$ edges of the form



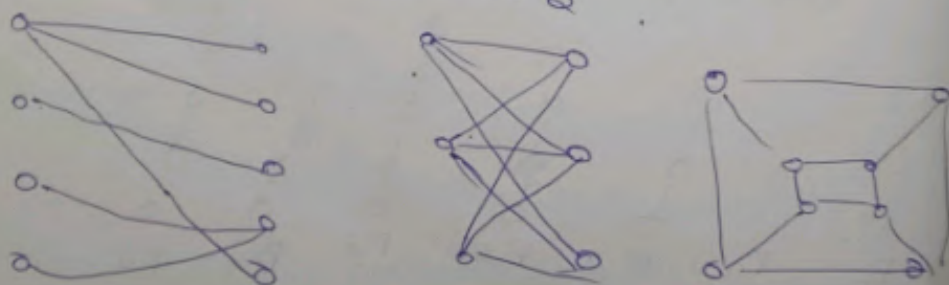
> Cycle (C_n): A cycle is a closed path in which all edges and all intermediate vertices are distinct.



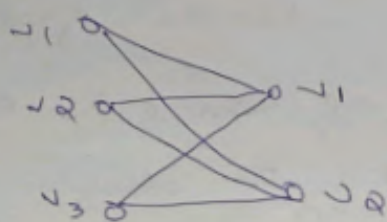
> Wheel graph (W_n): A wheel graph W_n is formed by connecting a single vertex to all vertices of a cycle C_n .



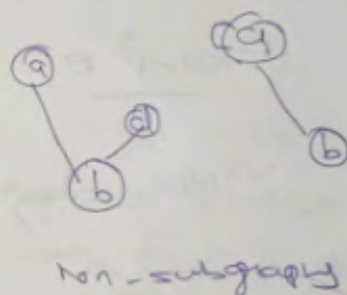
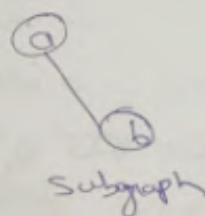
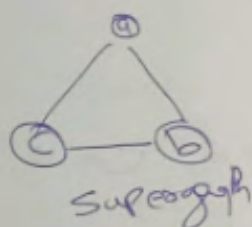
Bipartite Graph: A graph is bipartite if its vertex set can be divided into two disjoint sets V_1 and V_2 such that every edge connects a vertex in V_1 to one in V_2 .



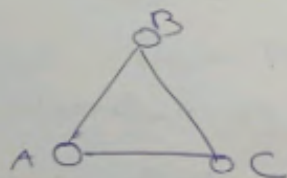
complete Bipartite Graph : A complete bipartite graph $K_{m,n}$ is a bipartite graph where every vertex in V_1 is connected to every vertex in V_2 .



Subgraph : A subgraph ~~of a~~ graph $G = (V, E)$ is a graph $H = (V', E')$ such that $V' \subseteq V$ and $E' \subseteq E$.



Is the following graph bipartite?

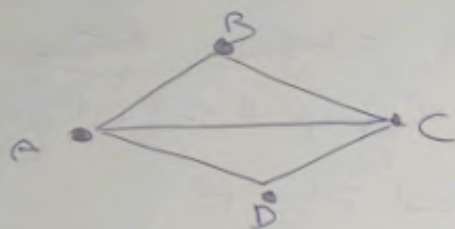


Answer : No it is not bipartite. A cycle of length 3 (odd length) cannot be bipartite because we cannot split the

vertices into two disjoint sets without
moving vertices within the same set.

HAND SHAKING LEMMA

Statement: In any undirected graph, the sum
of the degrees of all vertices is equal to
twice the number of edges.



Example:

Vertex degrees: $A=2, B=2, C=2, D=2$
Total = 8.

number of edges = 4.

$$\text{Hence, } 2|E| = \sum \deg(v) = 8.$$

Proof: Each edge in an undirected graph
contributes exactly 2 to the total degree
count (one for each endpoint), so the
sum of degrees equals $2|E|$.

Theorem: Even Number of odd Degree Vertices

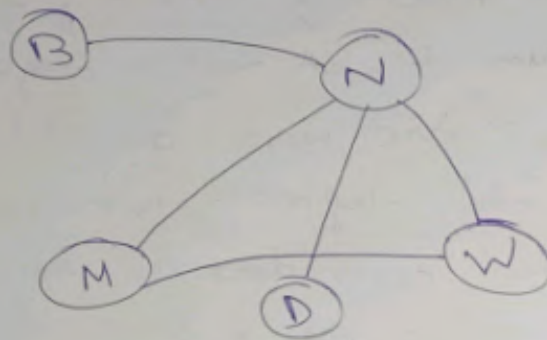
Statement: In any undirected graph the number of vertices with odd degree is even.

Proof: From the Handshaking Lemma, we know the total sum of all vertex degrees is even. Suppose there were an odd number of vertices with odd degrees; their sum would be odd, which contradicts the total being even. Hence, the number of odd-degree vertices must be even.

1) Do a graph model, stating the type of graph used, to represent airline routes where every day there are four flights from Boston to Newark, two flights from Newark to Boston, three flights from Newark to Miami, two flights from Miami to Newark, one flight from Newark to Detroit, two flights from Detroit to Newark, three flights from Newark to Washington, two flights from Washington to Newark and one flight from Washington to Miami, with

- a) An edge between vertices representing cities that have a flight between them (in one direction).

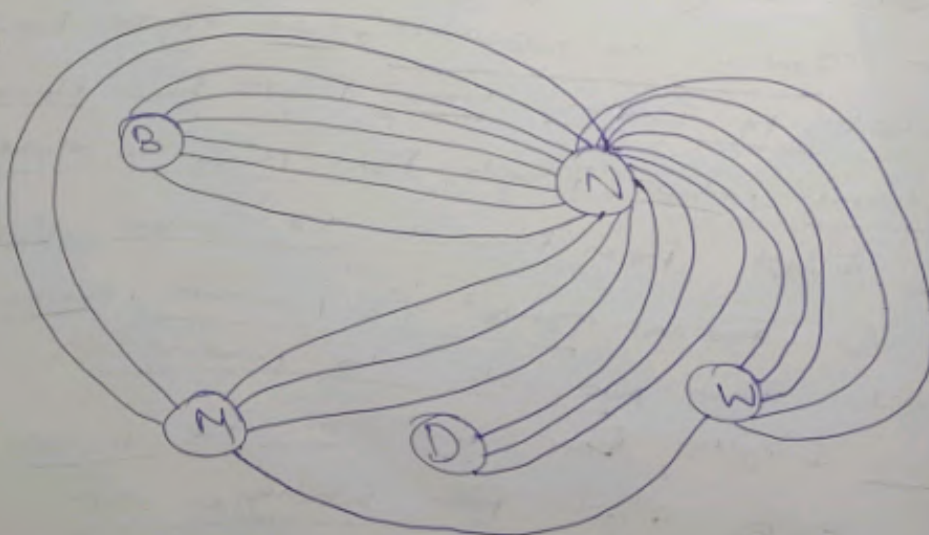
⇒



4: B → N
 2: N → B
 3: N → M
 2: M → N
 1: N → D
 2: D → N
 3: N → W
 2: W → N
 1: W → M

- b) An edge between vertices representing cities for each flight that operates between them (in either direction).

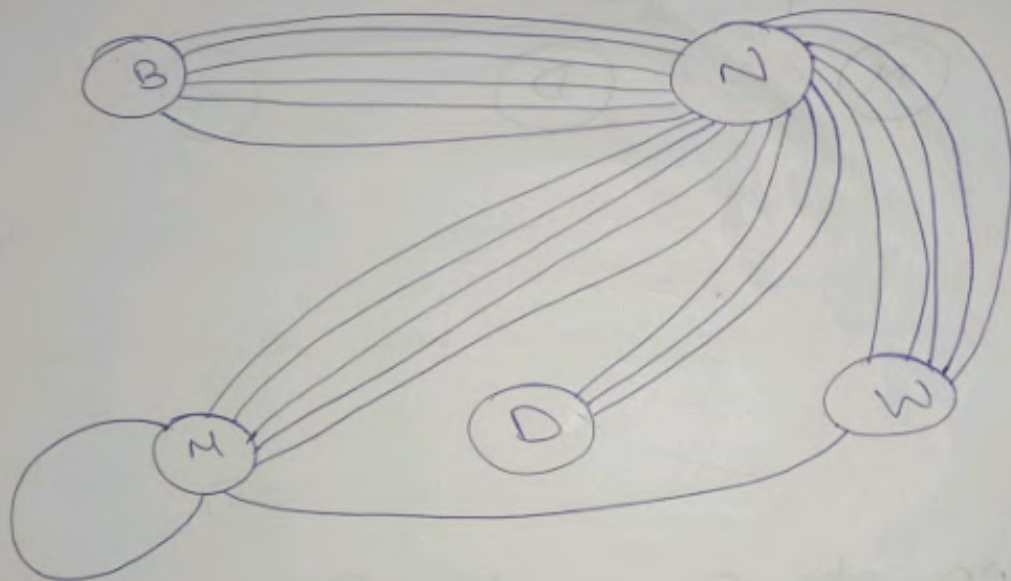
⇒



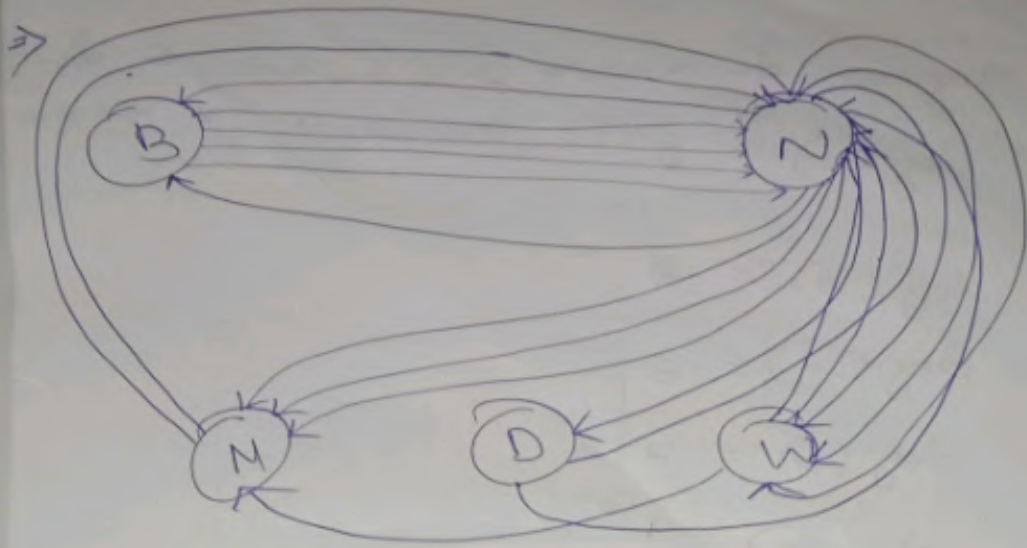
- d) An edge between vertices representing cities for each flight that operates between them (in either direction).

c) an edge between vertices representing cities for each flight that operates between them (in either direction), plus a loop for a special sightseeing trip that takes off and lands in Miami.

⇒



d) An edge from a vertex representing a city where a flight starts to a vertex representing the city where it ends.



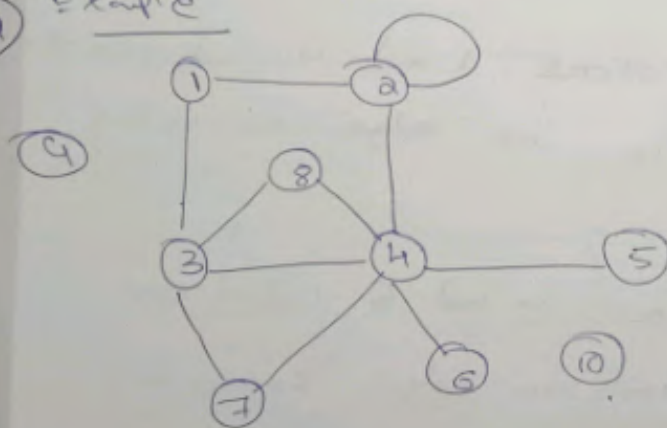
28/7/25

Does a graph contain isolated vertices?

⇒

(A)

2) Example



> vertices 9 and 10 are isolated vertices.

> vertices 6 and 5 are pendant vertices.

i) degree of vertex $\Rightarrow$ the number of edges incident on a vertex

$$\deg(1) = 2$$

$$\deg(2) = 2$$

$$\deg(4) = 6$$

$$\deg(3) = 4$$

$$\deg(5) = 1$$

$$\deg(6) = 1$$

$$\deg(7) = 2$$

$$\deg(8) = 2$$

degree of vertex (2) is 4. A loop
always adds 2 to the degree of the
vertex.

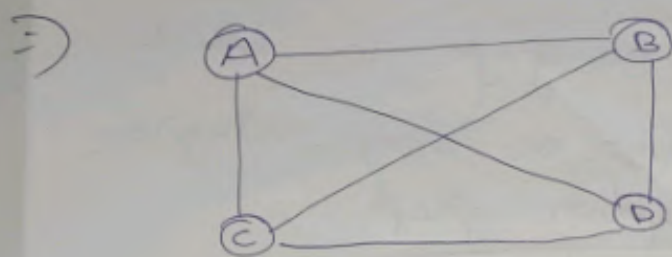
ii) Are the vertices 1 and 4 adjacent.

$\Rightarrow$ No, there is no edge connecting 1 and 4.

iii) Are the vertices 4 and 6 adjacent.

$\Rightarrow$ Yes, there is an edge connecting 4 and 6.

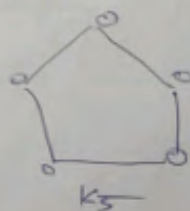
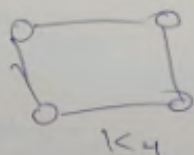
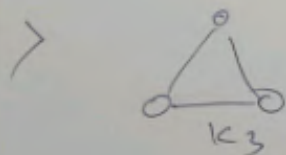
3) Can you draw a graph contain 4 vertices and any two pair have an edge connecting them?



complete graph $(K_n) \Rightarrow$ A graph with n vertices, such that any pair of vertices has an edge connecting them is called a complete graph.

> K_1 is the smallest graph possible and it is trivial graph.

> K_2 has two vertices and one vertex.



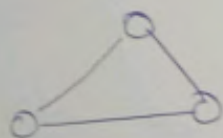
h) Can you draw a graph in which every vertex is of the same degree?

⇒ If we have a graph in which every vertex has a same degree is called regular graph.

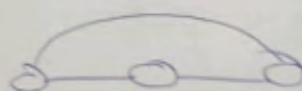
In Particular if a graph has vertices with degree k . Then the graph is called k -regular graph.

eg: regular graph.

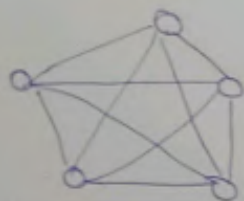
2 regular



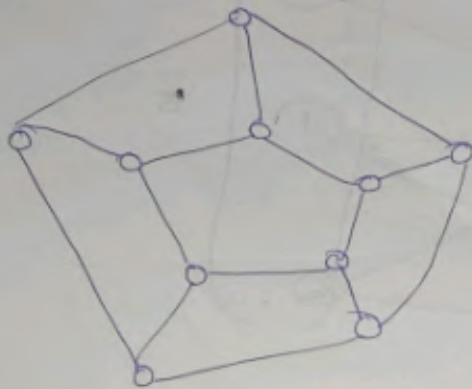
3 regular



k -regular.

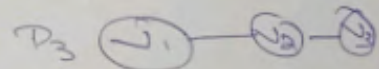
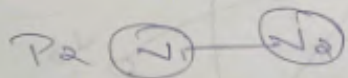
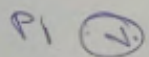


~~A graph which~~
 is a 3 regular graph which
 are not complete.

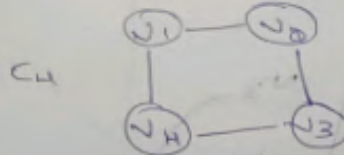
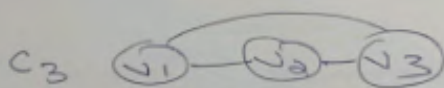


Classes of Graphs

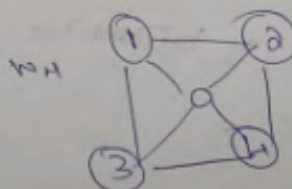
① Paths (P_n)



② Cycles (C_n)

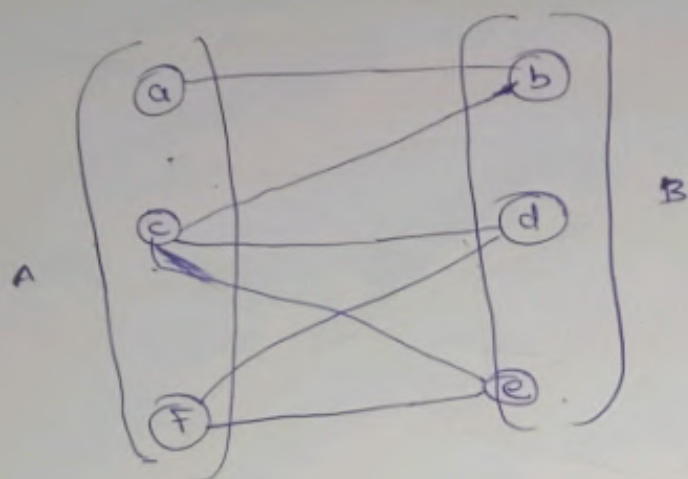


③ Wheels (W_n)

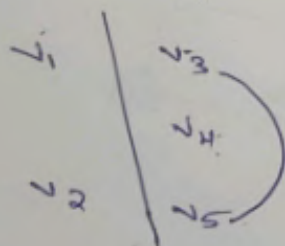
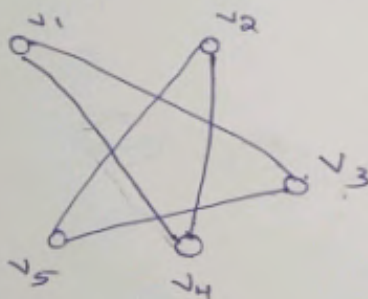


3/11/25

Bipartite Graph

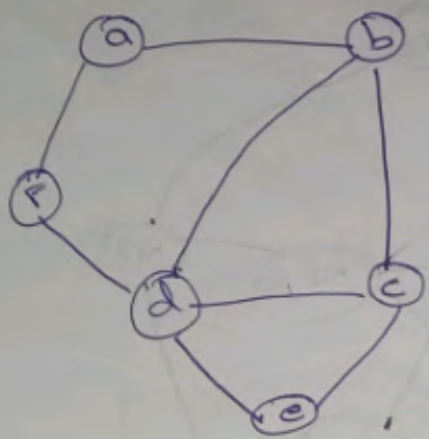


1) check if the following graphs are bipartite

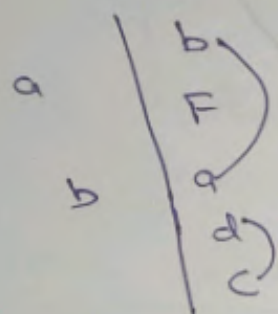


This is not a bipartite graph.

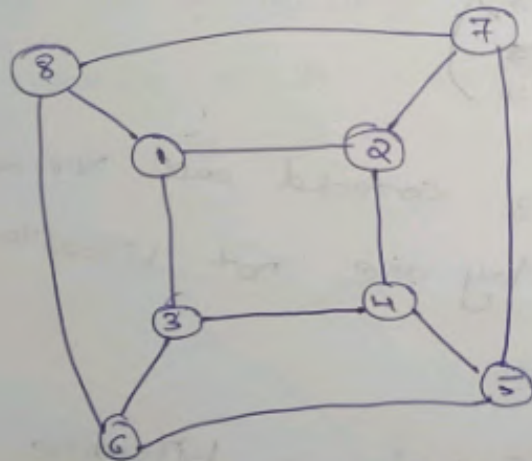
2)



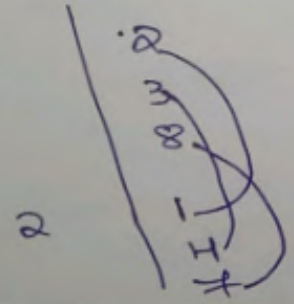
Let A and B be the partitions of the vertex set. If vertex c belong to A then clearly b, d and e should belong to B. Since there are edges connecting b and d and c this is not bipartite graph.

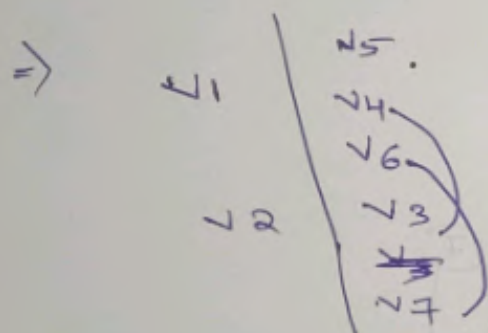
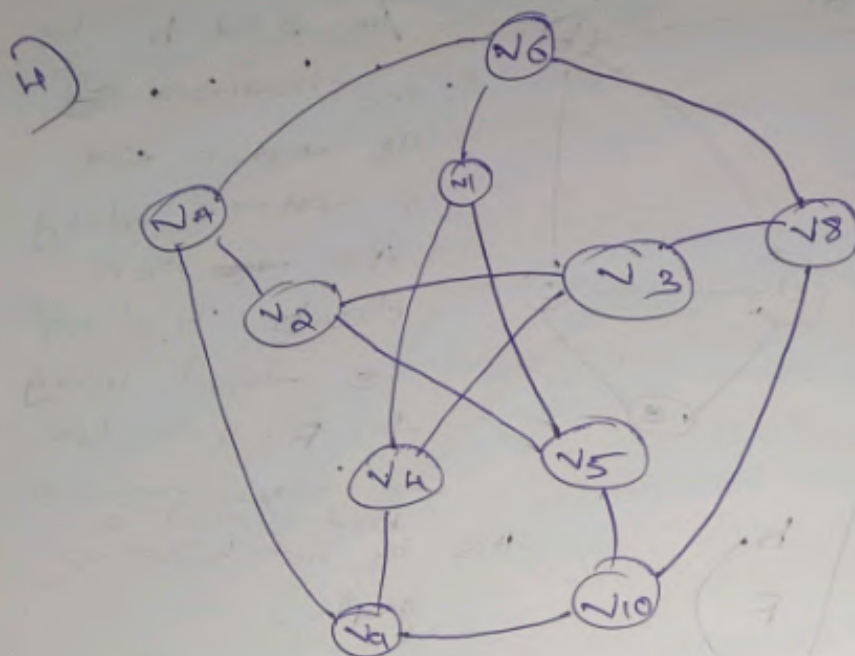


3)



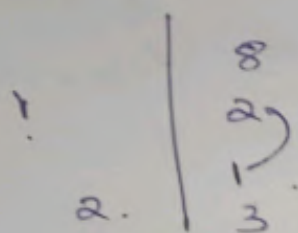
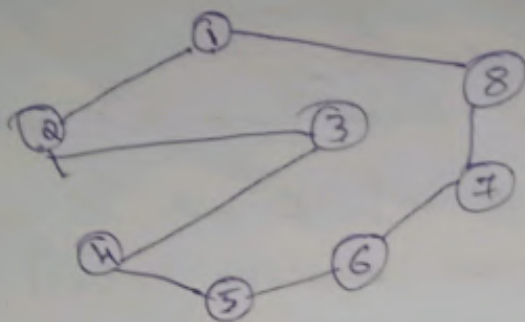
Since there is an edge connecting 2 and 1, 3 and 4 and 8 and 7 so it is not a bipartite graph.





since v_4 and v_3 connected and v_6 and v_7 are connected $\therefore$ they are not bipartite graph.

$\Rightarrow$ Check if G is a bipartite graph.

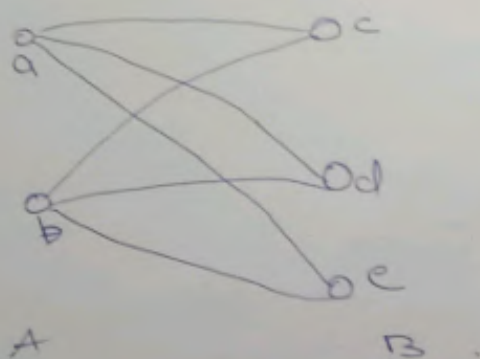


there is a connection
between 2 and 1 so it is
not bipartite graph.

1/8/25

Complete Bipartite

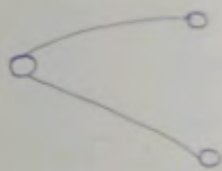
A complete bipartite graph with bipartition
 A and B is a bipartite graph in which
every vertex of A is adjacent to every
vertex of B and vice versa.



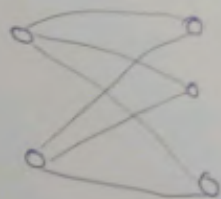
Note

IF G is a complete bipartite graph with bipartitions A and B such that A has m vertices and B has n vertices then the graph can be represented as $K_{m,n}$.

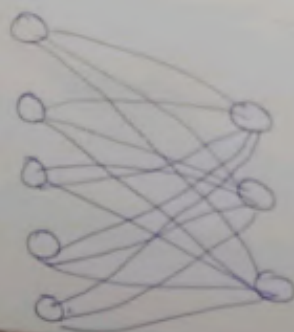
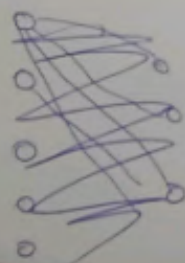
1) $K_{1,2}$



2) $K_{2,3}$



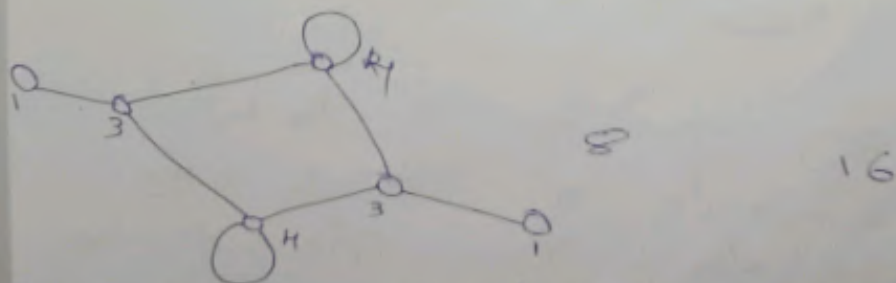
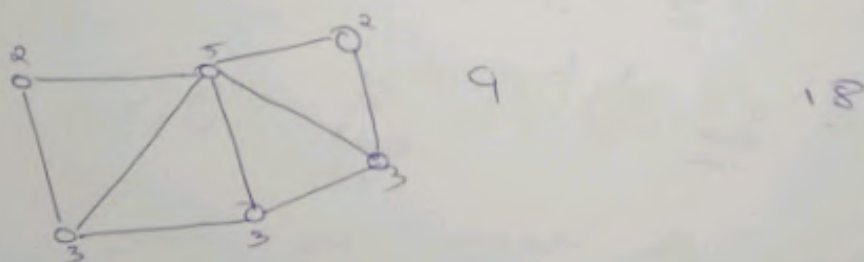
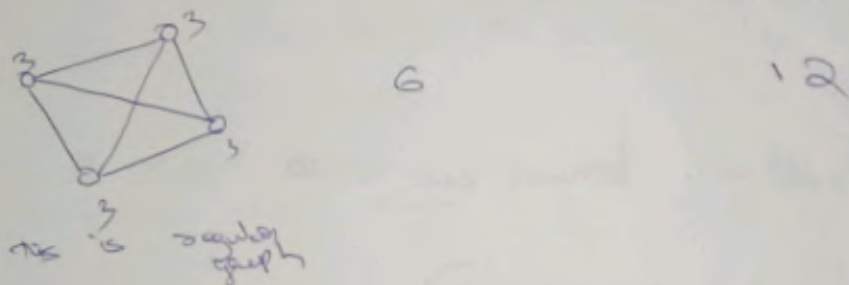
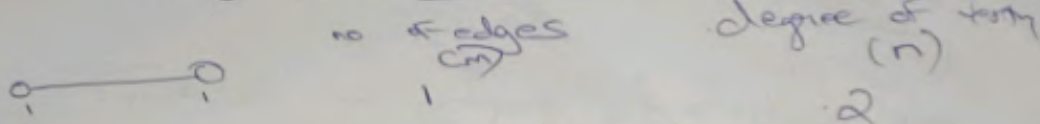
3) $K_{3,5}$



Handshaking Lemma

For any undirected graph sum of degrees of all the vertices is twice the number of edges of the graph.

Verify handshaking lemma in the following graph.



Here the sum of degree of vertex is double the no. of edges of graph.

2 × no of edges of graph = sum of degree of vertices.

Theorem

In any undirected graph the number of odd vertices will always be even.

Proof

from handshake lemma we know $2 \times M = \sum_{\text{vertices}} \deg(v)$

$$2m = \sum \deg(v)$$

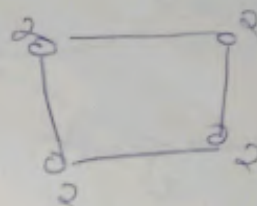
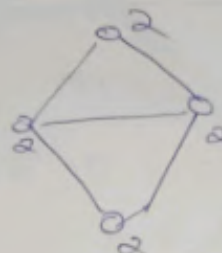
$$2m = \sum_{\text{even}} \deg(v) + \sum_{\text{odd}} \deg(v)$$

rearranging the terms we have.

$$2m - \sum_{\text{even}} \deg(v) = \sum_{\text{odd}} \deg(v)$$

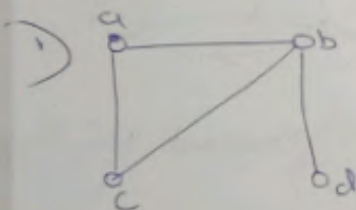
Since every term on the LHS is a multiple of two the RHS should also be the multiple of two.
Since every term on the RHS

is an odd number, For RHS
to be even we require even number
of terms on the RHS. This implies
there will be even number of odd
vertices.



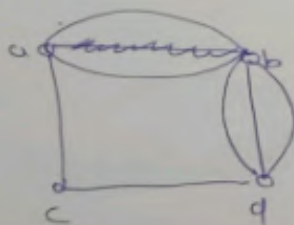
Q10

Determine the type of graph.



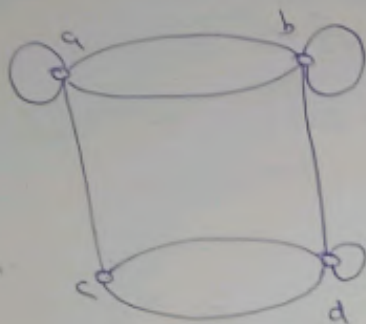
undirected
simple graph.

2)



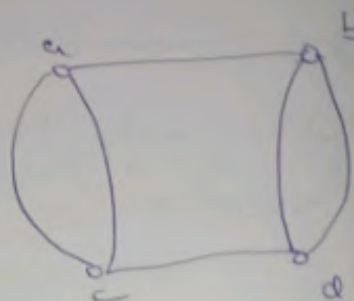
multigraph

3)



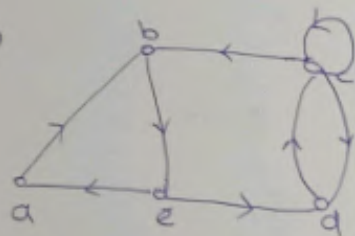
Pseudo graph.

4)



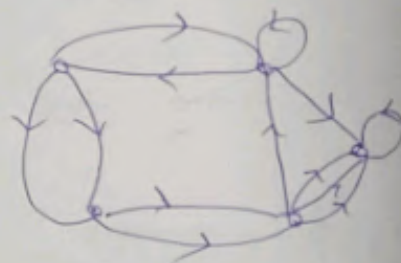
Multigraph

5)



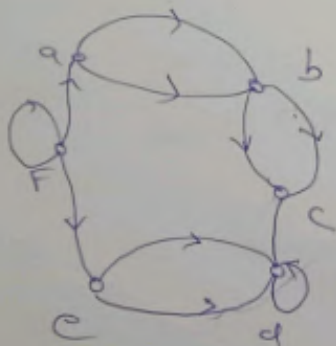
Pseudograph (directed)

6)



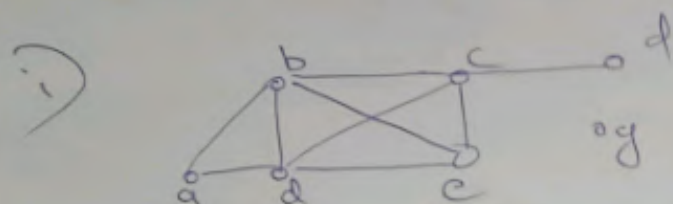
Directed Pseudograph

7)



Directed Pseudograph

What are the degrees and neighbourhoods of the vertices in the following graph.



$$A: \deg(a) = 2 : b, e$$

$$\deg(b) = 4 : a, f, e, c$$

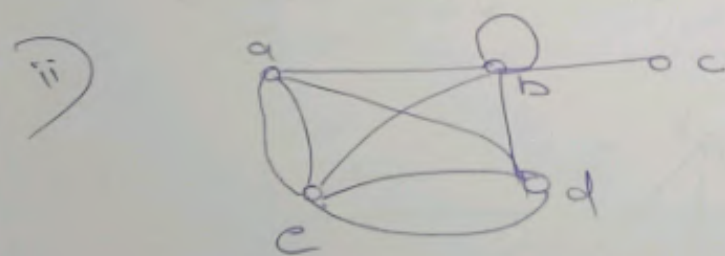
$$\deg(c) = 5 : b, f, e, d$$

$$\deg(d) = 1 : c$$

$$\deg(e) = 3 : c, b, f$$

$$\deg(f) = 4 : a, b, c, e$$

$$\deg(g) = 0$$



$$A: \deg(a) = 3 : b, e, d$$

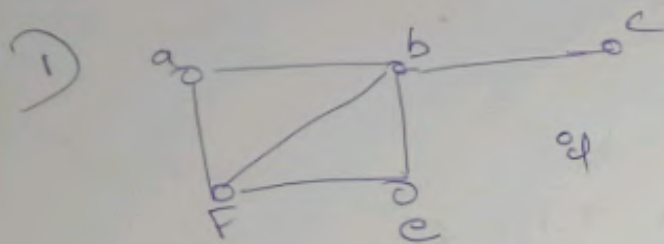
$$\deg(b) = 5 : c, d, e, a$$

$$\deg(c) = 1 : b$$

$$\deg(d) = 3 : b, e, a$$

$$\deg(e) = 4 : b, d, a$$

3) In the following graphs, find the no of vertices, edges and degrees of each vertex. identify all isolated and pendant vertices.



A: Vertices : 6

edges : 6

Pendant vertex : c

Isolated vertex : d

$$\deg(a) = 2$$

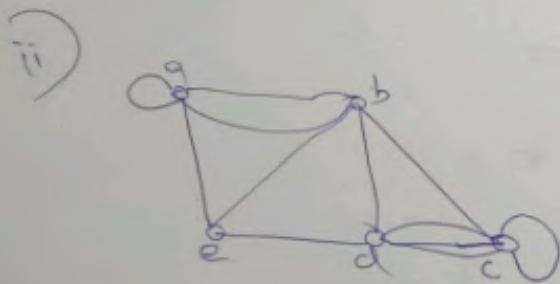
$$\deg(b) = 3$$

$$\deg(c) = 1$$

$$\deg(d) = 0$$

$$\deg(e) = 2$$

$$\deg(f) = 3$$



vertices = 5

edges = 11

$$\deg(e) = 6$$

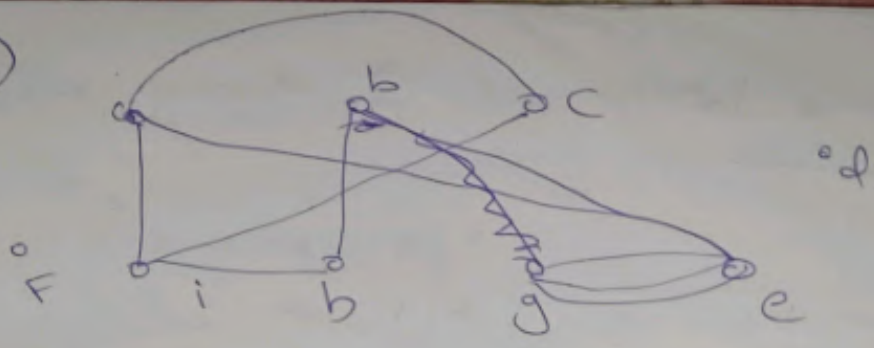
$$\deg(a) = 5$$

$$\deg(b) = 6$$

$$\deg(c) = 6$$

$$\deg(d) = 5$$

iii)



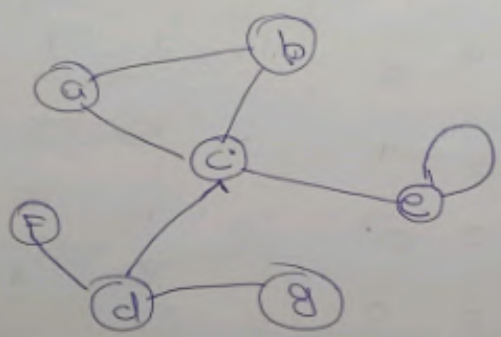
A: Vertices : 9
 edges = 12
 isolated vertices = f, d

$\deg(a) = 3$
 $\deg(b) = 2$
 $\deg(c) = 4$
 $\deg(d) = 0$
 $\deg(e) = 6$
 $\deg(f) = 0$
 $\deg(g) = 2$
 $\deg(h) = 3$
 $\deg(i) = 4$

H/8/25

Representation of a graph

Adjacency



① adjacency list

vertices	
a	b, c
b	a, c
c	a, b, c, d
d	f, c, g
e	c, e
f	d
g	d

② adjacency matrix

	a	b	c	d	e	f	g
a	0	1	1	0	0	0	0
b	1	0	1	0	0	0	0
c	1	1	0	1	1	0	0
d	0	0	1	0	0	1	0
e	0	0	1	0	1	0	0
f	0	0	0	1	0	0	0
g	0	0	0	1	0	0	0

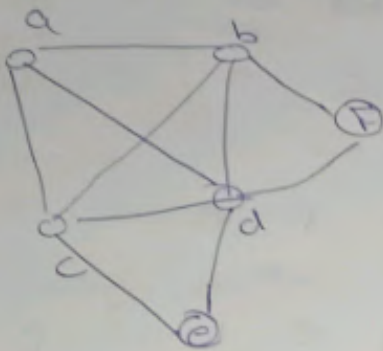
not incidence & multiple edges

③ incidence matrix

	ab	ac	bc	ce	cd	df	dg
a	1	1	0	0	0	0	0
b	1	0	1	0	0	0	0
c	0	1	1	1	1	0	0
d	0	0	0	0	1	1	1
e	0	0	0	1	0	0	0
f	0	0	0	0	0	1	0
g	0	0	0	0	0	0	1

no loops constraint

2) Draw adjacency list, adjacency matrix and incidence matrix.



adjacency list

Vertices	
a	b, c, d
b	a, d, F, c
c	a, b, d, e
d	a, b, c, F, e
e	c, d
F	b, d

adjacency matrix

	a	b	c	d	e	F
a	0	1	1	1	0	0
b	1	0	1	1	0	1
c	1	1	0	1	1	0
d	1	1	1	0	1	1
e	0	0	1	1	0	0
F	0	1	0	1	0	0

~~incidence matrix~~

~~cb ac ad bd bc~~

incidence matrix

	fb	ab	ac	ad	bc	bd	cd	de	df	ec
a	0	1	1	1	0	0	0	0	0	0
b	1	1	0	0	1	1	0	0	0	0
c	0	0	1	0	1	0	1	0	0	1
d	0	0	0	1	0	1	1	1	1	0
e	0	0	0	0	0	0	0	1	0	1
f	1	0	0	0	0	0	0	0	1	0

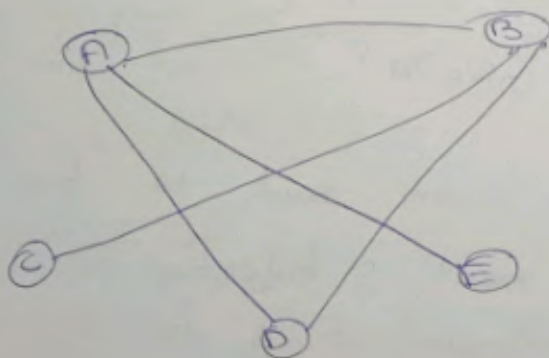
3) adjacency matrix

	a	b	c	d
a	0	1	1	0
b	1	0	0	1
c	1	0	0	1
d	0	1	1	0



incidence matrix

	AD	AB	BC	AE	BD
A	1	1	0	1	0
B	0	1	1	0	1
C	0	0	1	0	0
D	1	0	0	0	1
E	0	0	0	1	0



5/8/25

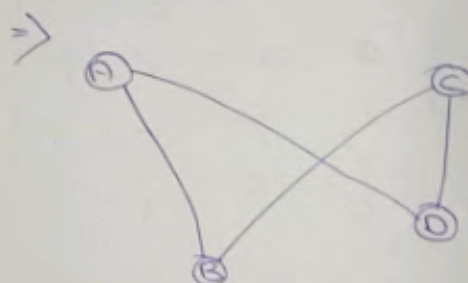
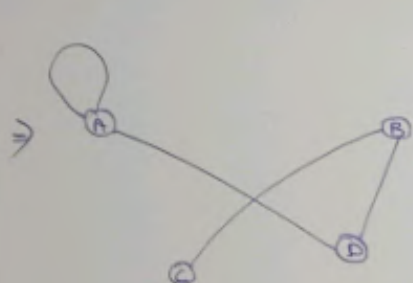
Draw the graph.

1) Adjacency matrix.

	AB	BA	C	D
A	1	0	0	1
B	0	0	1	1
C	0	1	0	0
D	1	1	0	0

2) Incidence matrix

	AB	BC	CD	AD
A	1	0	0	1
B	1	1	0	0
C	0	1	1	0
D	0	0	1	1



ISOMORPHISM GRAPH

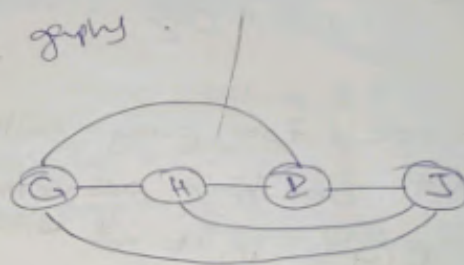
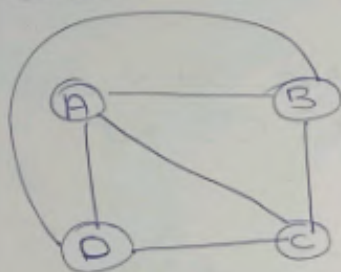
Two graphs g and h are said to be isomorphic if there exist a bijective function F from the vertex set of g to the vertex set of h such that whenever U and V are adjacent in g then $F(U)$ and $F(V)$ are adjacent in h .

and vice versa.

NOTE: If G and H are isomorphic graphs then they must have the same number of vertices, the same number of edges and ~~the~~ this degree sequence (list of degrees of each vertex) must be identical.

example 1

Consider the following graphs.



Graph G. Since the number of vertices and edges of G and H are the same and degree sequence of both the graphs is $(3, 2, 2, 2)$ there is a possibility that the graphs are isomorphic. let the bijective function from $V(G)$ to $V(H)$

be as follows

$$F(A) = G$$

$$F(B) = H$$

$$F(C) = I$$

$$F(D) = J$$

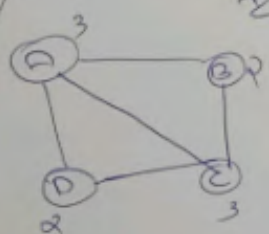
UV	$F(u)$	$F(v)$	Present in H
			✓
AB		GH	✓
BC		HZ	✓
BD		HJ	✓
CD		ZJ	✓
AD		GJ	
AC		GZ	✓

Since for every adjacent vertices u and v in G , we have adjacent vertices $F(u)$ and $F(v)$ in H and vice versa.

G and H are isomorphic.

Q25

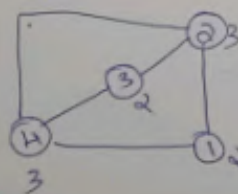
Check if the graph are isomorphic



no. of vertices = 4

no. of edges = 5

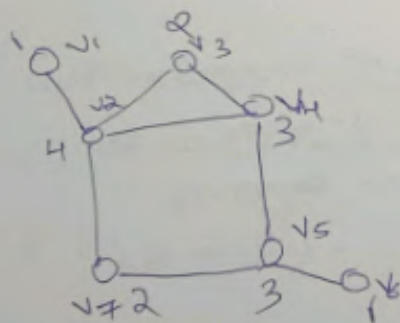
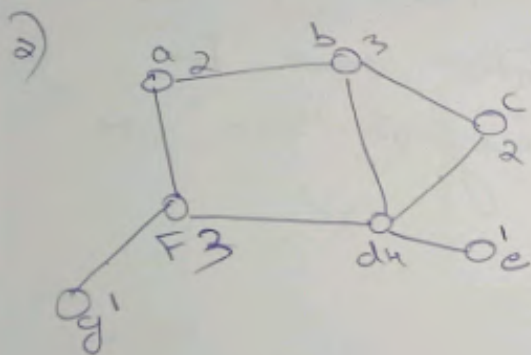
degree sequence $(3, 2, 3, 2)$



A	✓	3	1
B	✓	2	1
C	✓	3	1
D	✓	2	1

uv	$F(u) \cdot F(v)$	Present in H ?
AB	23	✓
BC	34	✓
CD	41	✓
DA	21	✓
AC	24	✓

∴ F is the bijective function that satisfies the condition described in the definition of isomorphic graph. Hence the graphs are isomorphic.



no. of vertices = ~~8~~ 7
 no. of edges = ~~8~~ 8
 degree sequence = $(2, 3, 2, 1, 4, 3, 1)$

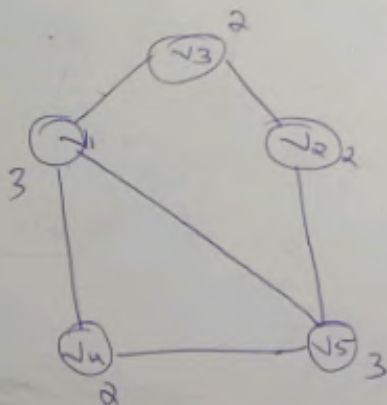
	F
a	✓3, ✓4
b	✓3, ✓4
c	✓3, ✓4
d	✓2
e	✓2
f	✓3
g	✓1

correction ✓
 so no adjacency

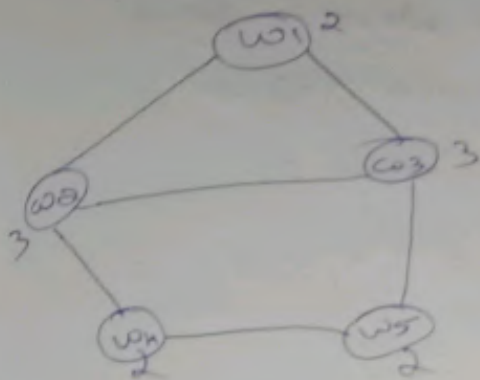
Since vertex ~~v_1~~ and ~~v_2~~ are the
 only vertices of degree 4 in G and H
 respectively $F(v_1)$ should be v_2 . Now
 since a is the vertex of degree 2
 it can be mapped to either v_3 or
 v_7 but note that since a and b
 are not adjacent $F(a) = v_3$ and $F(b)$ should
 not be adjacent. But both v_3 and v_7 are
 adjacent to v_2 . Hence a bijective function
 satisfying condition described in the
 definition of isomorphism cannot be
 constructed. Hence the graphs are not
 isomorphic.

8/8/25

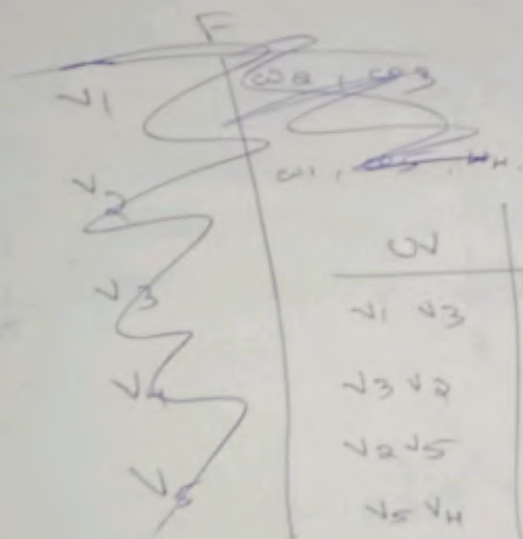
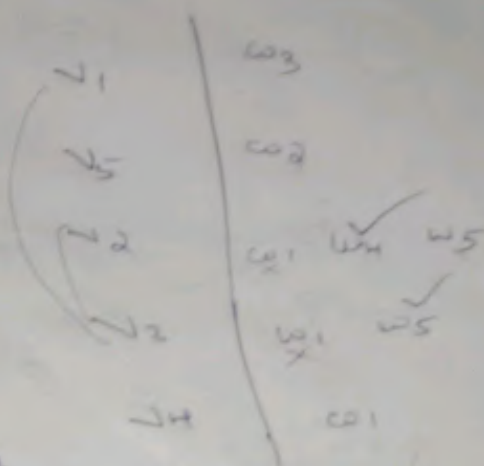
1)



no. of vertices = 5
 no. of edges = 6
 degree sequence
 $= (2, 2, 2, 3, 3)$



~~deg~~

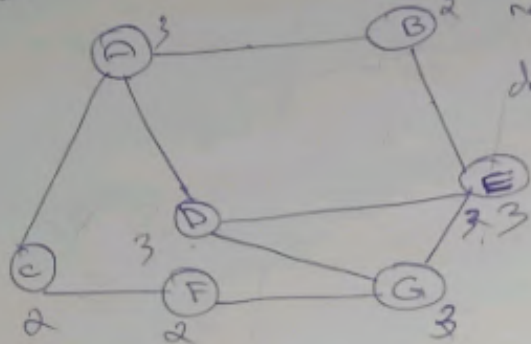


~~deg~~

w	$F(w), F(v)$	
v_1, v_3	w_3, w_5	✓
v_3, v_2	w_5, w_4	✓
v_2, v_5	w_4, w_2	✓
v_5, v_4	w_2, w_1	✓
v_4, v_1	w_1, w_3	✓
v_1, v_5	w_3, w_2	✓

It satisfies the bijective function together the graphs are bijective function.

2)

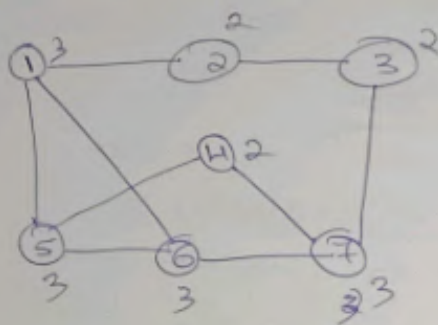


no. of edges $\Rightarrow 9$

no. of vertices $\Rightarrow 7$

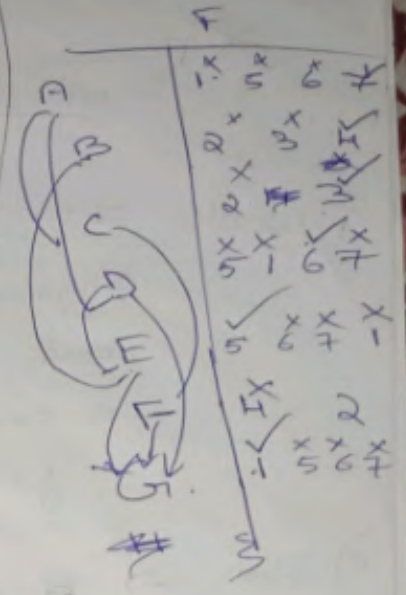
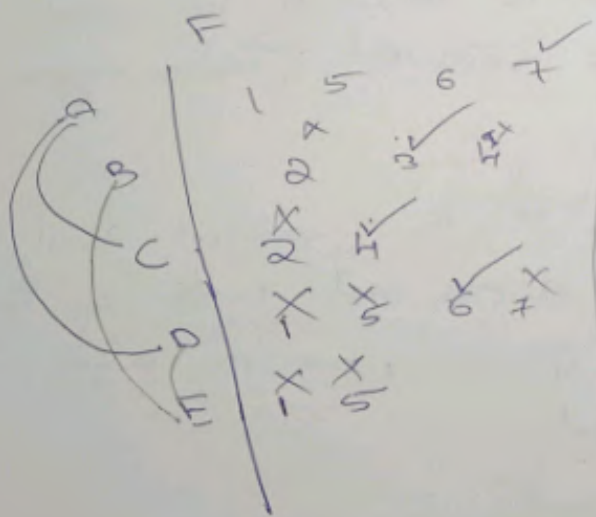
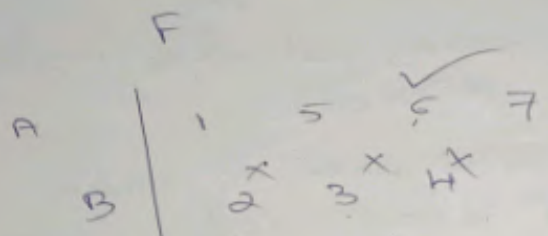
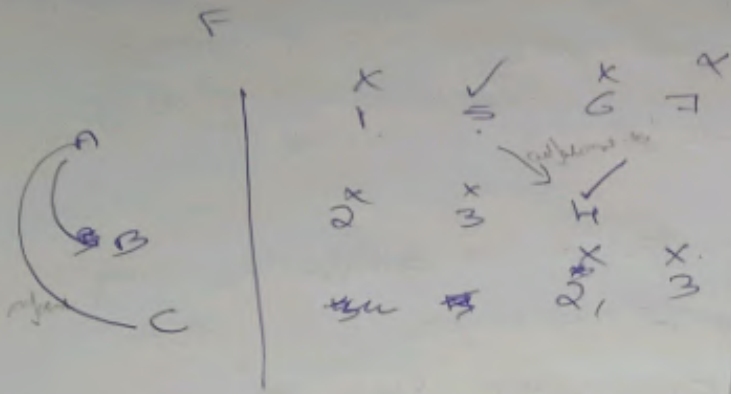
degree same $\Rightarrow$

(3, 3, 3, 3, 2, 2, 2)



if left side
is adjacent then
right is adjacent
(4 not adjacent
to 1)

	F			
	1	5	6	7
(deg 3) A	✓			
(deg 2) B	✓	✓	✓	
(deg 2) C	✓	✓		
D				
E				
F				
G				

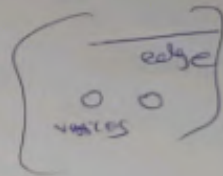


UV	F(U)	F(V)
AB	4	4
AC	3	3
AD	7	7
AE	5	5
AF		
AG		
BC		
BD		
DE		
BE		
BG		
CD		
CE		
CF		
CG		
DE		
DF		
DG		
EF		
EG		
FG		

11/8/25



CONNECTIVITY

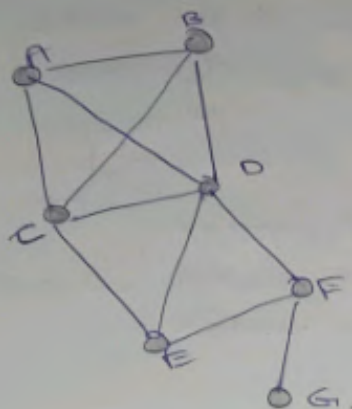


1) Walk $\rightarrow$ A walk in a graph is any sequence of vertices such that each consecutive pair of vertices is connected by an edge in the graph. (vertices and edges may be repeated).

2) Path $\rightarrow$ A path is a walk in which no vertex is repeated. (naturally edges are not repeated).

3) Trail $\rightarrow$ A trail is a walk in which no edges are repeated (but vertices may be repeated).

4) Circuit $\rightarrow$ A circuit is a closed trail which starts and ends at the same vertex, but no edges are repeated.



$ABDFECADFG \rightarrow \text{walk}$

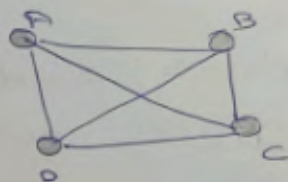
$ABDEF G \rightarrow \text{Path}$

$ABD A C D F \rightarrow \text{trail}$

$ABDCA \rightarrow \text{Circuit}$

5) Hamiltonian Path $\rightarrow$ A Path that visits each vertex exactly once is called a Hamiltonian Path.
(the edges can never be repeated). every vertex should be included.

6) Hamiltonian Circuit $\rightarrow$ A Hamiltonian Path that is closed - It starts and ends at the same vertex, visiting every vertex exactly once in between.

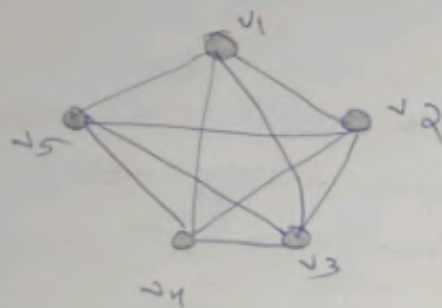


$ABCD \rightarrow \text{Hamiltonian Path}$

$ABCD A \rightarrow \text{Hamiltonian Circuit}$

7) Euler trail $\rightarrow$ A trail that uses every edge exactly once.

8) Euler circuit $\rightarrow$ An ^{trail that} eulerian is closed - starts and ends at the same vertex.



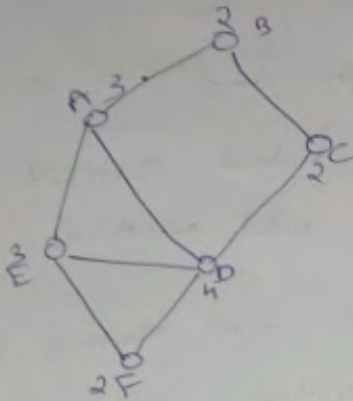
euler circuit $\rightarrow$ ~~$v_1 v_2 v_3 v_4 v_5 v_2 v_4 v_1$~~
 $v_1 v_2 v_3 v_4 v_5 v_2 v_4 v_1 v_3 v_5 v_1$

*
 9) Hamiltonian graph $\rightarrow$ A graph is said to be hamiltonian if it contains a hamiltonian circuit.

10) ~~Regular~~ Euler graph $\rightarrow$ A graph is said to be euler graph if it contains an eulers circuit.

14/1/25

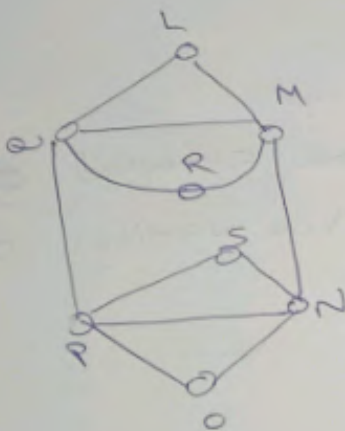
1)



It is not an Euler graph
because it has two odd degree vertices.

2)

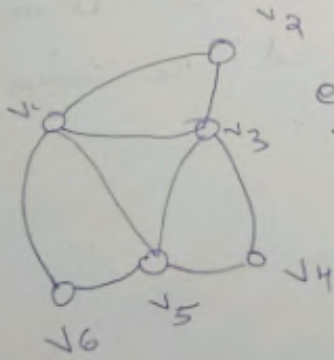
2)



LMNOPSNPQRNQL

~~LMNOPSNPQRNQL~~

3)



Euler circuit is possible
because all even degree vertices.

Theorems:

All vertices of the graph are .

if the graph contains an Euler circuit .

A graph has exactly two odd vertices .

A graph has exactly two odd vertices if and only if the graph contains an Euler trail .

⇒ Connected graph .

A graph is said to be connected if there connecting any two vertices of the graph .

Weighted graph

A graph is said to be weighted graph if the edges are assigned costs/lengths .

Dijkstra's

Step 1 : Construct a table with columns corresponding to every vertex of the graph .

Step 2 : In the first row assign the value zero to the source vertex (origin) and infinity to all other vertices.

Step 3 : To fill the next row look at all the vertices adjacent to the current vertex. update the columns corresponding to its neighbours with the dual edge weight from the source. If smaller than the existing value.

Step 4 : From the next row choose the vertex with the smallest value that has not been permanently marked.

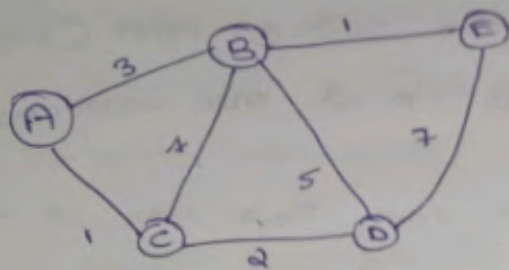
Step 5 : Construct this next row by relaxing all neighbours of S_i . For each neighbour update its column with the smallest of following two values.

The sum of the smallest value from the previous row and the edge weight connecting S_i to that neighbour.

The existing ^{entry} in that column.

Repeat steps 4 and 5 until shortest distance from the destination vertex is determined or until all vertices are permanently marked.

18/8/05, Find the shortest path from A to E.



⇒

	A	B	C	D	E
A (0)	0	∞	1	∞	∞
B	3	0	4	∞	1
C	1	4	0	2	5
D	∞	∞	2	0	7
E	∞	1	5	7	0

⇒

C(1) $\begin{pmatrix} 0 \\ 1, 1 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 3, 4 \end{pmatrix}$ 1 $\begin{pmatrix} 3 \\ 4, 2+1 \end{pmatrix}$ ∞

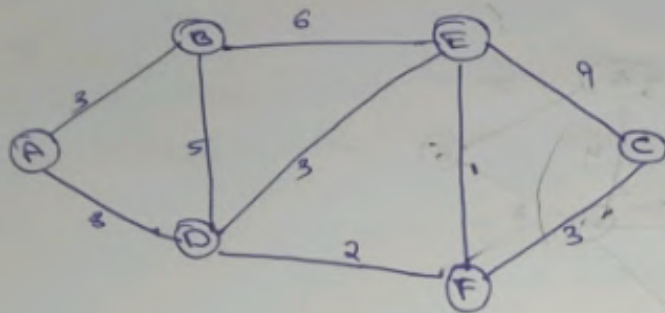
B(3) 0 3 $\begin{pmatrix} 1 \\ 1, 4 \end{pmatrix}$ $\begin{pmatrix} 3 \\ 3, 3+2 \end{pmatrix}$ $\begin{pmatrix} 4 \\ \infty, 4 \end{pmatrix}$

D(3) 0 $\begin{pmatrix} 3 \\ 3, 5+3 \end{pmatrix}$ 1 $\begin{pmatrix} 1 \\ 1, 2+3 \end{pmatrix}$ 3 $\begin{pmatrix} 4 \\ 4, 7+3 \end{pmatrix}$

E(4) 0 $\begin{pmatrix} 3 \\ 3, 4+1 \end{pmatrix}$ 1 $\begin{pmatrix} 3 \\ 3, 7+1 \end{pmatrix}$ 4

Shortest distance is 4//.

2) Find the shortest distance from A to, all other vertices of the graph.



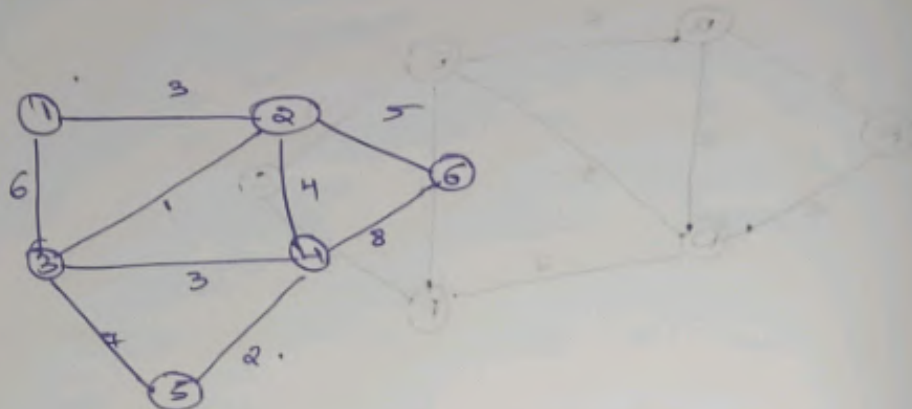
⇒

	A	B	C	D	E	F
A	0	∞	∞	∞	∞	∞
B(3)	0	3	∞	8	9	∞
D(8)	0	3	∞	8	9	10
E(9)	0	3	18 (∞, 18)	8 (8, 13)	9	10
F(10)	0	3	13 (8, 13)	8 (8, 12)	9 (9, 11)	10
C(13)	0	3	13	8	9 (9, 29)	10 (19, 6)

Shortest distances.

Shortest distance from A to B = 3, A to C = 13, A to D = 8, A to E = 9 and A to F = 10, A to A = 0.

- 2) Find the shortest distance from vertex 1 to all other vertices of the graph. Also, mention the path to be followed to reach there.



⇒

	1	2	3	4	5	6
1	0	3	6	∞	∞	∞
2(3)	0	3	4	7	∞	8
3(4)	0	3	4	7	11	8
4(7)	0	3	4	7	9	8
6(8)	0	3	4	7	9	8
5(9)	0	3	4	7	9	8

the shortest distance from vertex 1 to 2 = 3
 1 to 3 = 6, 1 to 4 = 7, 1 to 5 = 9, 1 to 6 = 8

Path

1 (3)

1 (3)

1 (3)

1 (3)

1 (3)

19/8/25

Travel

The T

a new

Coast

STEPS

Step 1

Step 2

Step 3

Step 4

Step 5

Path

1 (3) 2

1 (3) 2 (11) 3

1 (3) 2 (4) 4

1 (3) 2 (4) 4 (5) 5

1 (3) 2 (5) 6

19/8/25

Travelling salesman Problem (TSP)

The TSP is the Problem of ~~constructing~~ finding a hamiltonian cycle/circuit of minimal total cost in a weighted graph.

STEPS

Step 1 : choose the starting vertex as the one with maximum degree. mark it as visited.

Step 2 : From the current vertex look at all the edges leading to vertices that are not yet visited.

Step 3 : select the edge with minimal weight among them.

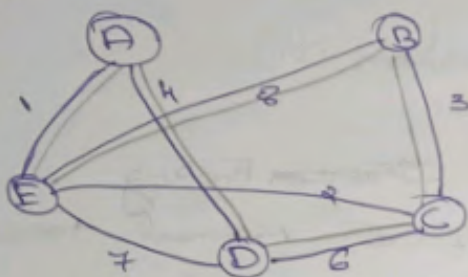
Step 4 : move to that vertex and mark it as visited.

Step 5 : repeat the steps 2 to 4 until all vertices have been visited exactly once.

Step 6 : Return to the starting vertex
by choosing the edge from the
last visited vertex ~~back to~~
~~the start~~

Step 7 : Sum up all the chosen edge weights
to find the total travel cost.

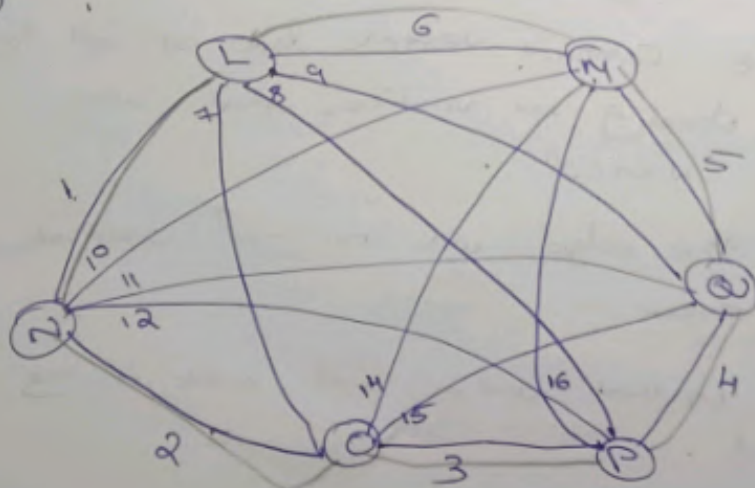
1)



~~Path~~ $E \xrightarrow{1} A \xrightarrow{4} D \xrightarrow{6} C \xrightarrow{3} B \xrightarrow{8} E$

$$1 + 4 + 6 + 3 + 8 = 22 //$$

2)

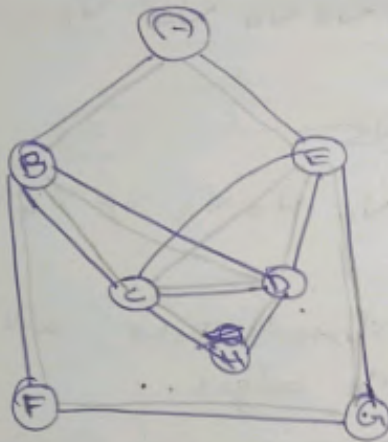


$$L \xrightarrow{1} N \xrightarrow{2} O \xrightarrow{3} P \xrightarrow{4} Q \xrightarrow{5} M \xrightarrow{6} L$$

$$1+2+3+4+5+6 = 21 //$$

Chinese Postman Problem (CPP)

Solve CPP for the following ~~graph~~



ABFGEA

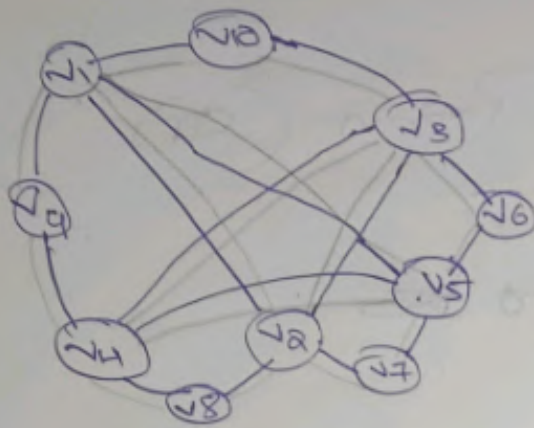
ECHDE

ABFGIECHDEA

CDBC

ABFGIECDBCCHDEA

2)



V1 V10 V3 V6 V5 V7 V2 V8 V4 V9 V1

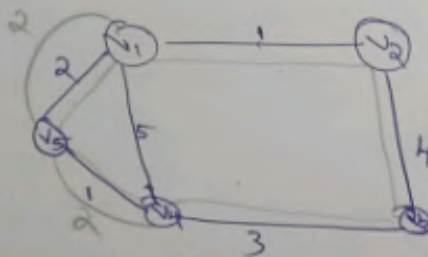
~~V1 V10~~

V5 V4 V3 V2 V1 V5

V1 V10 V3 V6 V5 V4 V3 V2 V1 V5 V7 V2 V8 V4 V9 V1

NOTE: Length of the above shortest path will be sum of weights of all the edges...

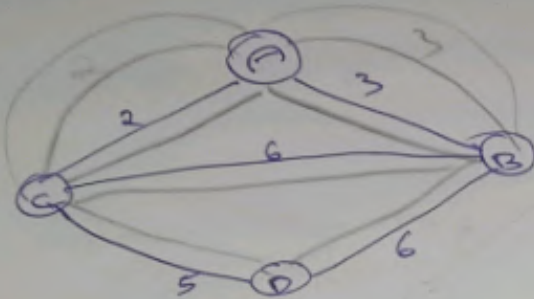
3)



V1 V5 V3 ^{Short} → 3

~~V1 V2 V3 V4 V5~~
~~V1 V2 V3 V4 V5~~
~~V1 V2 V3 V4 V5~~
~~V1 V2 V3 V4 V5~~

3)



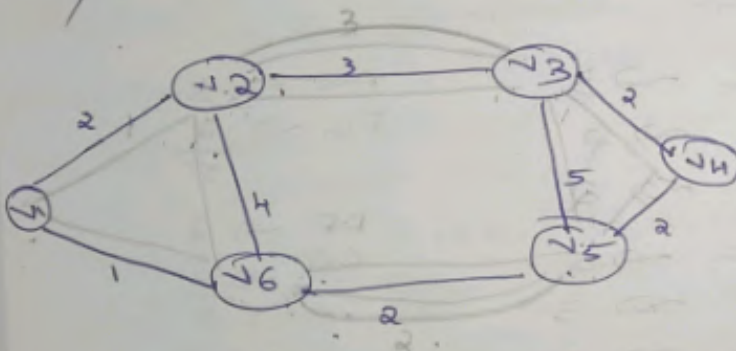
Find all possible
shortest paths
duplicate

ABDCA

~~ABDBCA~~

ABCA BDCA

26/8/25 Save the CPP for the following



$$V2V5 \rightarrow V3V6 \quad \times$$

$$5 + 6 = 11$$

$$V2V6 \rightarrow V3V5 \quad \times$$

$$3 + 4 = 7$$

adj list $\{V2, V3, V5, V6\}$

$$V2V3 \rightarrow V2V3 \rightarrow 3$$

$$V2V5 \rightarrow V2V6V5 \rightarrow 7$$

$$V2V6 \rightarrow V2V6 \rightarrow 3$$

$$V3V5 \rightarrow V3V4V5 \rightarrow 4$$

$$V3V6 \rightarrow V3V4V5V6 \rightarrow 6$$

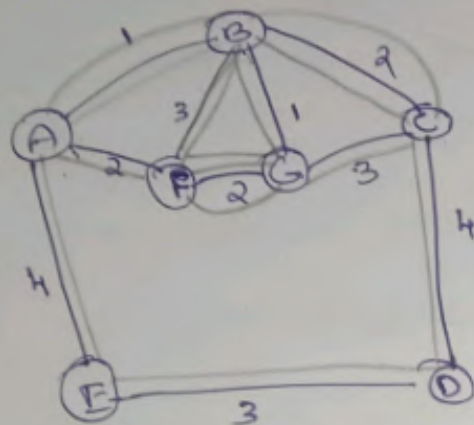
$$V5V6 \rightarrow V5V6 \rightarrow 2$$

$$V1V2V3V4V5V6V1$$

$$V1V2V3V5V6V2V3V4V5V6V1 = 26$$

Find the shortest path and cost
✓ $V6V6 \rightarrow 2$ choose the
 $V2V3 \rightarrow 3$ + one more
5/ the shortest
path with minimum

2)



$\{A, C, F, G\}$

$AC \rightarrow ABC \rightarrow 3$
 $AF \rightarrow AF \rightarrow 2$
 $AG \rightarrow ABG \rightarrow 2$
 $CF \rightarrow CGF \rightarrow 5$
 $CG \rightarrow CG \rightarrow 3$
 $FG \rightarrow FG \rightarrow 2$

$AC \rightarrow 3$
 $FG \rightarrow \frac{2}{5} //$ ✓

$AF \rightarrow 2$
 $CG \rightarrow \frac{3}{5} //$

$AG \rightarrow 2$
 $CF \rightarrow \frac{5}{7} //$

Shortest Path =

A B C D E

A B C G F A B C D E

A B C G F A B F G B C D E

= $\frac{30}{26} //$

2/2/22

Module 2

Counting Techniques

Addition Rule

If an event can happen in 'm' ways, and another different event can happen in 'n' ways, then the total number of ways to choose one of the events is $m+n$ ways (the events must be disjoint). eg. ~~the events are~~

A student wants to choose a subject where he has three different choices in mathematics and 4 different choices in science. Here the student has a total $3+4=7$ choices. Since choosing math ~~is~~ ^{OR} science are separate choices.

Multiplication Rule

If a task can be done in 'm' ways and for each of these another task can be done in 'n' different ways, then both the tasks can be done together in $m \times n$ ways

example: ~~for~~ if a meal consists of two choices of starters and three choices of main dishes there are $2 \times 3 = 6$ meal choices (since a choice has to be made from starters and main dishes).

algas

Permutation.

A Permutation is an arrangement of objects in order, if order matters, we use Permutation.

No. of ways to arrange ~~the~~ objects from n distinct objects is given by the formula

$${}^n P_r = P(n, r) = \frac{n!}{(n-r)!}$$

where $n! = 1 \times 2 \times 3 \times \dots \times n$.

eg: no. of ways in which the letters of word BEST can be used to form words with 3 letters is given by ${}^4 P_3 = \frac{4!}{(4-3)!}$

$$= \frac{4!}{1!} = \frac{1 \times 2 \times 3 \times 4}{1} = 24$$

BES	BET	EST	SBE	SET	TSE
BSF	BTF	ETS	SEB	STF	TBS
BST	ESB	BBT	STB	TBE	TES
BTS	ESB	ETB	SBT	TBT	TSE

COMBINATIONS

A combination is selection of objects without caring its order of selection. If order does not matter we use combination.

The number of ways to choose r objects from n distinct objects is given by ~~nCr~~

$$nC_r = {}^n C_r = \frac{n!}{(n-r)! \cdot r!}$$

eg:

Two students from a class of 5 students can be chosen in ${}^5 C_2 = \frac{5!}{(5-2)! \cdot 2!}$

$$\begin{aligned} &= \frac{5!}{3! \cdot 2!} = \frac{1 \times 2 \times 3 \times 4 \times 5}{(1 \times 2 \times 3) \cdot (1 \times 2)} \\ &= \frac{20}{2} = 10 // \end{aligned}$$

Q. If A, B, C, D and E are the 5 students of the class. Two students could be chosen in one of the following ways.

AB BC CD DE
AC BD CE
AD BE
AE

1) A student can choose to read either a maths book (with 4 choices) or an english book (with 5 choices). How many different choices ~~has~~^{does} the student have.

$\Rightarrow$ using addition theorem no of choices = $4+5 = 9$.

2) In How many ways can 5 students sit in a row of 3 chairs.

$$\Rightarrow {}^5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = \frac{1 \times 2 \times 3 \times 4 \times 5}{1 \times 2} = 60$$

3) How many different 4 digit numbers can be formed using the digits 1, 2, 3, 4, 5; if repetition of digits is not allowed

$$\Rightarrow {}^5P_4 = \frac{5!}{(5-4)!} = \frac{1 \times 2 \times 3 \times 4 \times 5}{1} = 120$$

4) How many different arrangements of letters of the word MATHS are possible?

$$\Rightarrow {}^5P_5 = \frac{5!}{(5-5)!} = \frac{120}{1} = 120$$

8/9/25

5) From a group of 10 players, how many ways can a team of 4 players be selected?

6) A school ID Number consists of two letters followed by three digits. How many different ID numbers are possible? (where character repetition is not allowed)

7) A committee of two Boys and three Girls is to be formed from 5 Boys and 6 girls. How many such committees can be formed?

8) In how many ways can the first 3 prizes be awarded to 10 competitors?

9) A box contains 10 red, 8 blue, and 6 green balls. we have to select 2 balls, how many ways can you select these, if the colour doesn't matter. also In how many ways can you do this selection if both the balls are of the same colour.

⇒ 5) $n = 10, r = 4$

$$nCr = \frac{n!}{(n-r)! r!}$$

$$10C4 = \frac{10!}{(10-4)! 4!}$$

$$= \frac{10}{6! 4!}$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(1 \times 2 \times 3 \times 4) \times (1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \times 9 \times 10)} = \frac{5040}{24 \times 5040} = 210$$

variation of each number 7. no
 generate the school no numbers & changes
 have to be done in a second drawing
 can be changed in 26 different ways

when ~~replaced~~
 1

8) $n=10, r=3$

$${}^{10}P_3 = \frac{10!}{(10-3)!} = \frac{10 \times 9 \times 8}{1} = 720$$

9) $\&$

12 bills whose color not significant
 04 C2 ways.

R, R, ... B1, B2, B3, ... G1, G2, ... G3.

2 red 2 blue 2 green
 10C2 8C2 6C2.

$$10C2 + 8C2 + 6C2 = 45 + 28 + 15 = 88$$

11/10/15

What is the minimum number of students that should be present in a class, to guarantee that at least 2 students achieve the same score ⁱⁿ the final exam, if the exam is ~~graded~~ on a scale from 0 to 100?

2) How many people should be present in a class such that there must be at least 2 people with the same birthday?

3) How many words must be written such that there must be at least 2 words that begin with the same letter?

⇒ 101 → 102 students

2) 366 → 367 people

B) 26 words but ^{we need} 27 words:
26 → 27 words

Pigeon hole Principle

If case of positive integers and $\boxed{K+1}$ one or more objects are placed into K boxes then there is atleast one box containing two or more of the objects. $\rightarrow$ box should be at least 2 objects box

1) How many strings of length are can be formed from the english alphabets.

2) A group of 30 people have been trained as astronauts to go on the first mission to Mars. How many ways are there to select a crew of 6 people to go on this mission.

3) In how many ways can you select 5 cars from a store which has 52 cars.

4) Let $A = \{A, B, C, D\}$ How many subsets of order two are there in the set A .

5) Suppose the salesman has to visit 5 different cities such that he must begin his trip in the specified city

$$\Rightarrow \overset{\text{series no}}{n} P \underset{\text{ways}}{\neq}$$

Recurrence relation

1) What is recurrence relation?

A recurrence relation is an equation that defines each term of a seq. as a function of its previous term. It refers to an term of a seq. in term of earlier terms.

eg: $a(n)$, $a(n-1)$... any term is obtained by adding 2 to the previous term.

2) General form of a Recurrence Relation

General form: $a(n) = F(a(n-1))$

where $a(n) = n^{\text{th}}$ $a(n-k)$, $\dots$, $a(n-k)$
 Function, $k = \text{order of the recurrence}$, $F = \text{some}$

3) Solution of a Recurrence Relation

A solution is a formula that gives the value $a(n)$ directly without recursion.

eg. $a(n) = a(n-1) + 2$ with $a(0) = 1$ has solution $a(n) = 2n + 1$.

4) method 1: Iteration (substitution) method

Steps:

1. Write the recurrence and initial condition.

eg. $a(n) = a(n-1) + 2, a(0) = 1$.

2. expand step by step:

$$a(n) = a(n-1) + 2 = (a(n-2) + 2) + 2 = \dots = a(n-k) + 2k$$

after k steps: $a(n) = a(n-k) + 2k$.

3. Stop when you reach the initial condition

$$a(0) = 1. \text{ as } a(n) = 1 + 2n.$$

• example (Haoabri method)

Recurrence : $a(n) = a(n-1) + 5$, $a(0) = 4$

Soln : $a(n) = 4 + 5n$

Recurrence : $a(n) = 2a(n-1)$, $a(0) = 1$

Solution : $a(n) = \cancel{2n} 2^n$

5) mod 2 : characteristic eqn method

used for linear recurrence relations with constant coefficients.

general form : $a(n) = c_1 + a(n-1) + c_2 \cdot a(n-2)$
($c_1, c_2 \in \mathbb{R}$)

steps

1. write recurrence in standard form.

2. form the characteristic polynomial by using $a(n) = x^n$

3. ~~Factorize~~ solve for x (roots of polynomial)

4. write general solution depending on type of roots

→ use initial conditions to solve for constants

Case 1: Distinct Roots

If roots are distinct $r_1, r_2, \dots, r_k$ then

$$a(n) = A_1 \cdot r_1^n + A_2 \cdot r_2^n + \dots + A_k \cdot r_k^n$$

eg: $a(n) = 3a(n-1) + 4a(n-2), a(0)=1, a(1)=2$

character eqn $r^2 - 3r - 4 = 0 \Rightarrow$ roots $4, -1$

genl soln $a(n) = A \cdot 4^n + B \cdot (-1)^n$

on condns $\Rightarrow A = 8/5, B = 3/5$

Final solution $a(n) = (8/5) \cdot 4^n + (3/5) \cdot (-1)^n$

Case 2: Repeated roots

If a root r has

10/9/25

Find the first 5 terms of the sequence defined by each of the following recurrence relations and initial conditions.

i) $a_n = 6a_{n-1}$, where $a_0 = 2$.

$$a_1 = 6 \times a_0 = 6 \times 2 = 12 //$$

$$a_2 = 6 \times a_1 = 6 \times 12 = 72 //$$

$$a_3 = 6 \times a_2 = 6 \times 72 = 432 //$$

$$a_4 = 6 \times a_3 = 6 \times 432 = 2592 //$$

$$a_5 = 6 \times a_4 = 6 \times 2592 = \underline{\underline{15552}}$$

ii) $a_n = a_{n-1}^2$; $a_1 = 2$.

$$a_1 = 2$$

$$a_2 = a_1^2 = 4 //$$

$$a_3 = a_2^2 = 16 //$$

$$a_4 = a_3^2 = 256 //$$

$$a_5 = a_4^2 = 256^2 = \underline{\underline{65536}}$$

iii) $a_n = a_{n-1} + 3a_{n-2}$,
where $a_0 = 1, a_1 = 2$

$$\begin{aligned} a_2 &= a_1 + 3a_0 \\ &= 2 + 3 \\ &= 5 // \end{aligned}$$

$$\begin{aligned}
 a_3 &= a_2 + 3a_1 \\
 &= 5 + 3 \times 2 \\
 &= 5 + 6 = 11 //
 \end{aligned}$$

$$\begin{aligned}
 a_4 &= a_3 + 3a_2 \\
 &= 11 + 3 \times 5 \\
 &= 11 + 15 \\
 &= \underline{\underline{26}}
 \end{aligned}$$

$$\begin{aligned}
 a_5 &= a_4 + 3a_3 \\
 &= 26 + 3 \times 11 \\
 &= 26 + 33 \\
 &= \underline{\underline{59}}
 \end{aligned}$$

$\rightarrow$
 i) $a_n = n a_{n-1} + n^2 a_{n-2}$
 $a_0 = 1 \quad a_1 = 1$

$$\begin{aligned}
 a_2 &= 2 \times a_1 + 2^2 \times a_0 \\
 &= 2 \times 1 + 4 \times 1 \\
 &= 6 //
 \end{aligned}$$

$$\begin{aligned}
 a_3 &= 3 \times a_2 + 3^2 \times a_1 \\
 &= 3 \times 6 + 9 \times 1 \\
 &= 18 + 9 \\
 &= 27 //
 \end{aligned}$$

$$a_4 = 4 \times a_3 + 4^2 \times a_2 = 4 \times 27 + 16 \times 6 = 204 //$$

$$a_5 = 5 \times a_4 + 5^2 \times a_3 = 5 \times 204 + 25 \times 27 = 1020 + 675 = 1695 //$$

$$2) \quad a_n = a_{n-1} + a_{n-3}$$

$$a_0 = 1, \quad a_1 = 2, \quad a_2 = 0.$$

$$a_3 = a_2 + a_0$$

$$= 0 + 1 = 1 //$$

$$a_4 = a_3 + a_1$$

$$= 1 + 2 = 3 //$$

$$a_5 = a_4 + a_2$$

$$= 3 + 0 = 3 //$$

$$v) \text{ let } a_n = 2^n + 5 \times 3^n.$$

$$\text{for } n = 0, 1, 2, 3, \dots$$

$$i) \text{ Find } a_0, a_1, a_2, a_3 \text{ and } a_4.$$

$$b) \text{ show that } a_2 = 5a_1 + 6a_0,$$

$$a_3 = 5a_2 + 6a_1 \text{ and } a_4 = 5a_3 + 6a_2.$$

$$\Rightarrow a) \quad a_0 = 2^0 + 5 \times 3^0$$

$$= 1 + 5$$

$$= 6 //$$

$$a_1 = 2^1 + 5 \times 3^1$$

$$= 2 + 15$$

$$= 17 //$$

$$a_2 = 2^2 + 5 \times 3^2$$

$$= 4 + 5 \times 9 = 49 //$$

$$= 49 //$$

$$a_3 = 2^3 + 5 \times 3^3$$

$$= 8 + 135$$

$$= \underline{\underline{143}}$$

$$a_4 = 2^4 + 5 \times 3^4$$

$$= 16 + 5 \times 81$$

$$= 405 + 16$$

$$= \underline{\underline{421}}$$

12/9/25

b) $a_2 = 5a_1 - 6a_0$

LHS $a_2 = 49 \quad \text{--- (1)}$

RHS

$$5a_1 + 6a_0$$

$$= 5(17) + 6(6)$$

$$= 85 + 36 = \underline{\underline{121}}$$

$$\begin{pmatrix} a_1 = 17 \\ a_0 = 6 \\ a_2 = 49 \end{pmatrix}$$

$$a_3 = 5a_2 - 6a_1$$

LHS

$$a_3 = 143 //$$

--- (1)

RHS

$$5a_2 - 6a_1 = (5 \times 49) - (6 \times 17)$$

$$= 245 - 102 = \underline{\underline{143}} \quad \text{--- (2)}$$

$$a_n = 5a_3 - 6a_2$$

LHS

$$a_n = 421$$

RHS

$$5 \times 143 - 6 \times 29$$

$$= 715 - 294$$

$$= 421 //$$

iii) $a_n = 5a_{n-1} - 6a_{n-2}$

LHS

$$a_n = 2^n + 5 \cdot 3^n \quad (1)$$

RHS

$$5a_{n-1} - 6a_{n-2}$$

$$\begin{aligned} &= 5(2^{n-1} + 5 \cdot 3^{n-1}) - 6(2^{n-2} + 5 \cdot 3^{n-2}) \\ &= (5 \times 2^{n-1}) + (25 \times 3^{n-1}) - (6 \times 2^{n-2}) - (30 \times 3^{n-2}) \\ &= 5 \times 2^n \times 2^{-1} + (25 \times 3^n \times 3^{-1}) - (6 \times 2^n \cdot 2^{-2}) - \\ &\quad \cancel{30 \times 3^n \times 3^{-2}} \end{aligned}$$

$$\begin{aligned} a^{m+n} &= a^m \cdot a^n \\ a^{-m} &= \frac{1}{a^m} \end{aligned}$$

$$\begin{aligned}
 &= (5 \times 2^2 \times 2^1) + (25 \times 3^2 \times 3^1) - (6 \times 2^1 \times 2^2) - (30 \times 3^1 \times 3^2) \\
 &= (5 \times 2^2 \times \frac{1}{2}) + (\frac{25 \times 3^2 \times 1}{3}) - (6 \times 2^1 \times \frac{1}{4}) - (\frac{30 \times 3^1 \times 1}{9}) \\
 &= (5 \times 2^2 \times \frac{1}{2}) - (6 \times 2^1 \times \frac{1}{4}) + (\frac{25 \times 3^2 \times 1}{3}) - (\frac{30 \times 3^1 \times 1}{9}) \\
 &= 2^n \left[\frac{5}{2} - \frac{6}{4} \right] + 3^n \left[\frac{25}{3} - \frac{10}{3} \right] \\
 &= 2^n [1] + 3^n [5]
 \end{aligned}$$

$$= 2^n + 3^n \cdot 5 \quad \text{--- (2)}$$

(1) = (2) $\therefore$ LHS = RHS Hence Proved.

iv) Check if the sequence $a_n = 3n$ for every non-negative integer is a solution of the recurrence relation $a_n = 2a_{n-1} - a_{n-2}$ for $n=2, 3, 4$.

$$\Rightarrow \text{LHS : } a_n = 3n \quad \text{--- (1)}$$

$$\text{RHS : } 2a_{n-1} - a_{n-2}$$

$$a_n = 3n \\ \therefore a_{n-1} = 3(n-1)$$

$$2(3(n-1)) - 3(n-2)$$

$$a_{n-2} = 3(n-2)$$

$$= 12n - 12$$

$$6n - 6 - 3n + 6$$

$$= 3n \quad \text{--- (2)}$$

hence $\{a_n = 3n\}$ is a solution of the recurrence relation.

$$2) \quad \{a_n\} = \{2^n\} \quad \text{--- (1)}$$

$$a_n = 2a_{n-1} - a_{n-2} \quad \text{for } n \geq 2, 3, 4$$

$$a_n = 2[a_{n-1}] - [a_{n-2}]$$

$$= [4^n - 2] - [2^n - 2]$$

$$= 4^n - 2^n$$

$$= 2^n \quad \text{--- (2)}$$

LHS = RHS

Hence

$$\{a_n\} = \{2^n\}$$

is a solution of the recurrence relation

of the

$$> \quad 2(2^{n-1}) - 2^{n-2}$$

$$= 2(2^n \times 2^{-1}) - 2^n \times 2^{-2}$$

$$\begin{aligned} a_n &= 2^n \\ a_{n-1} &= 2^{n-1} \\ a_{n-2} &= 2^{n-2} \end{aligned}$$

$$2 \times 2^n \times \frac{1}{2} - 2^n \times \frac{1}{4}$$

$$2^n \left[1 - \frac{1}{4} \right] = \frac{3}{4} \cdot 2^n$$

LHS $\neq$ RHS

hence it is not a recursive soln.

19/9/25

1) check if the sequence $\{a_n\}$ is a solution of the recurrence relation ~~$a_n = -3a_{n-1} + 4a_{n-2}$~~

$$a_n = -3a_{n-1} + 4a_{n-2}$$

IF a) $a_n = 0$

b) $a_n = 1$

c) $a_n = (-4)^n$

d) $2(-4)^n + 3$

2)

LHS $a_n = 0$

RHS $-3a_{n-1} + 4a_{n-2}$

$$= -3(0) + 4(0) = 0 //$$

b) $a_n = 1$

RHS

$$-3a_{n-1} + 4a_{n-2}$$

$$= -3(1) + 4(1) = -3 + 4 = 1 //$$

$$c) \text{ LHS } a_n = (-4)^n. \quad \text{--- ①}$$

RHS

$$-3a_{n-1} + 4a_{n-2}$$

$$-3((-4)^{n-1}) + 4((-4)^{n-2})$$

$$= -3[(-4)^n (-4)^{-1}] + 4[(-4)^n (-4)^{-2}]$$

$$= -3\left[(-4)^n \cdot \frac{1}{(-4)}\right] + 4\left[(-4)^n \left(\frac{1}{(-4)^2}\right)\right]$$

$$= \frac{-3}{-4} (-4)^n + \frac{4}{16} (-4)^n$$

$$= (-4)^n \left[\frac{-3}{-4} + \frac{1}{4} \right] = 1$$

$$= (-4)^n \quad \text{--- ②}$$

$$\text{①} = \text{②}$$

LHS = RHS $\therefore$ the given solution is a recurrence relation.

d) $2(-H)^n + 3$

Solutions to recursive relation

Chapman's Equation

Linear Homogeneous Recurrence Relation with constant coefficients.

The general form of a 1st is given by
 $a_n = C_1 \cdot a_{n-1} + C_2 \cdot a_{n-2} + \dots + C_k \cdot a_{n-k}$

where $C_1, C_2, C_3, \dots, C_k$ are constants.

The recursive relation in the definition is the relation

i) linear because the right hand side is a sum of previous terms of the sequence

Can a multiple of it.

ii) Homogeneous because no terms occur that are not multiples of the previous terms of the sequence.

iii) The co-efficients of the terms of the sequence are all constants rather than functions that depend on n .

iv) Is a recurrence relation if a_n is expressed in terms of the previous k terms of the sequence. Then degree of the recurrence relation is said to be k .

→ Determine which of the following are linear homogeneous recurrence relations with constant co-efficients. Also find the degree of these ~~relations~~ ~~that~~.

i) $a_n = 3a_{n-1} + 4a_{n-2} + 5a_{n-3}$ ($k=3$)

ii) $a_n = 2n \cdot a_{n-1} + a_{n-2}$

iii) $a_n = a_{n-1} + a_{n-4}$

iv) $a_n = a_{n-1} + 2$

v) $a_n = a_{n-2}$

vi) $a_n = a_{n-1} + a_{n-2}$

vii) $a_n = a_{n-1} + n$

$$viii) a_n = 3a_{n-2}$$

$$ix) a_n = a_{n-1}^2$$

$$x) a_n = 3$$

$$xi) a_n = a_{n-1} + 2a_{n-3}$$

$$xii) a_n = \frac{a_{n-1}}{n}$$

$$xiii) a_n = a_{n-1} + a_{n-2} + n + 3$$

$$xiv) a_n = 4a_{n-2} + 5a_{n-4} + 9a_{n-7}$$

solve

(If linear
R.H.S. is 0 then
= 1)

$$i) a_n = a_{n-1} + a_{n-2} \text{ with } a_0 = 2, a_1 = 7$$

$$a_n = x^n$$

$$x^n = x^{n-1} + x^{n-2}$$

$$1 = 1/x + 1/x^2$$

$$x^2 = x + 1$$

$$x^2 - x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1 + 4}}{2}$$

$$= \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)}$$

$$= \frac{-1 \pm \sqrt{1 + 4}}{2}$$

$$= \frac{1+3}{2} = 2 \quad \checkmark$$

$$r_1 = 2, r_2 = -1$$

$$r_1, r_2, r_3, \dots$$

$$a_n = c_1(r_1)^n + c_2(r_2)^n + c_3(r_3)^n \dots$$

$$a_n = c_1(2)^n + c_2(-1)^n \quad \text{--- (1)}$$

$$a_0 = c_1 \cdot 1 + c_2 \cdot 1$$

$$\boxed{a = c_1 + c_2} \quad \text{--- (A)}$$

$$a_1 = c_1(2) + c_2(-1)$$

$$\boxed{7 = 2c_1 - c_2} \quad \text{--- (B)}$$

$$a = c_1 + c_2$$

$$7 = 2c_1 - c_2$$

$$\frac{a = c_1 + c_2}{7 = 2c_1 - c_2} \quad \text{--- (1)}$$

$$\textcircled{A} + \textcircled{B}$$

$$c_1 = 3 \quad \checkmark$$

$$\textcircled{A} = 2 > 3 + c_2$$

$$c_2 = -1 \quad \checkmark$$

$$a_n = \underline{\underline{3(2^n) - 1(-1)^n}}$$

$$2) \quad a_n = a_{n-1} + 6 \times a_{n-2} \quad \text{com } a_0 = 3, a_1 = 6.$$

$$a_n > r^n.$$

$$r^n > r^{n-1} + 6 \times r^{n-2}.$$

$$\frac{r^n}{r^n} = \frac{r^{n-1}}{r^n} + \frac{6r^{n-2}}{r^n}$$

$$1 = r^{-1} + 6r^{-2}$$

$$1 = \frac{1}{r} + 6 \cdot \frac{1}{r^2}.$$

$$r^2 = r + 6$$

$$r^2 - r - 6 = 0$$

$$r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times -6}}{2 \times 1}$$

$$= \frac{-1 \pm \sqrt{1+24}}{2}$$

$$= \frac{1+5}{2} = 3, -2.$$

$$a_n = C_1 (3)^n + C_2 (-2)^n \quad \text{--- (1)}$$

$$a_0 = 3$$

$$a_1 = 6$$

$$n=0 \quad \rightarrow \quad 3 = C_1 + C_2 \quad \text{--- (A)}$$

$$n=1 \quad \rightarrow \quad 6 = 3C_1 - 2C_2 \quad \text{--- (B)}$$

Soln ① and ②

$$\begin{aligned} a \times 2 &= 6 = 2c_1 + 2c_2 \\ b &= 6 = 3c_1 - 2c_2 \\ 1a &= 5c_1 \quad 0 \end{aligned}$$

$$c_1 = 12/5 //$$

$$3 = 12/5 + c_2$$

$$\begin{aligned} c_2 &= 3 - 12/5 \\ &= 3/5 \end{aligned}$$

Subst in ①

$$a_n = \frac{12}{5} (3)^n + \frac{3}{5} (-2)^n$$

Solve the recurrence relation

$$a_n = a_{n-1} - \frac{1}{4} a_{n-2}$$

where $a_0 = 1, a_1 = 2$

$$\Rightarrow i) \quad \sigma^n = \sigma^{n-1} - \frac{1}{4} \sigma^{n-2}$$

$$1 = \frac{1}{\sigma} - \frac{1}{4} \cdot \frac{1}{\sigma^2}$$

$$\sigma^2 = \sigma - \frac{1}{4}$$

$$\sigma^2 - \sigma + \frac{1}{4} = 0$$

$$4\sigma^2 - 4\sigma + 1 = 0$$

$$\text{Soln } \sigma = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{(-1) \pm \sqrt{x^2 - 4x + 4}}{2x4}$$

$$= \frac{1 \pm \sqrt{16 - 16}}{8} = \frac{1}{8} = 1/8$$

$$\underline{\underline{r = 1/2, 1/2}}$$

$$a_n = (c_1 + nc_2)(1/2)^n$$

$$n=0 \Rightarrow a_0 = c_1(1/2)^0 = c_1 \Rightarrow 1 = c_1$$

$$n=1 \Rightarrow a_1 = (c_1 + c_2)(1/2) = 2 \Rightarrow (1 + c_2)1/2 = 2$$

$$1 + c_2 = 4 \Rightarrow c_2 = 3$$

$\Rightarrow$ solve $a_n = -6a_{n-1} - 9a_{n-2}$
 IF $a_0 = 2, a_1 = 3$

$$\text{Ans : } a_n = x^n$$

$$x^n - 6x^{n-1} - 9x^{n-2}$$

$$1 = \frac{6x^{n-1}}{x^n} - \frac{9x^{n-2}}{x^n}$$

$$1 = \frac{-6}{x} - \frac{9}{x^2}$$

$$x^2 = -6x - 9$$

$$x^2 + 6x + 9 = 0$$

$$\text{soln } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times 9}}{2 \times 1}$$

$$= \frac{-6 \pm \sqrt{36 - 36}}{2}$$

$$= \frac{-6}{2}, -3, -3 //$$

$$a_n = (C_1 + n C_2) (-3)^n$$

$$n=0 \quad a_0 = (C_1) (-3)^0 = C_1$$

$$a_0 = 2 \quad C_1 = 2$$

$$n=1$$

$$a_1 = (C_1 + C_2) (-3) = (2 + C_2) (-3)$$

$$= 3 - 6 - 3C_2$$

$$9 = -3C_2$$

$$C_2 = -3 //$$

$$\therefore a_n = (2 - 3n) \underline{\underline{(-3)^n}}$$

Unit 3

$$A = \{a, b, c\}$$

Power set of A.

$$P(A) = \left\{ \{a, b, c\}, \phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\} \right\}$$

$$|A| = 3$$

$$|P(A)| = 2^3 = 8 //$$

2) List the members of these set.

i) $\{x : x \text{ is a real number} : x^2 = 1\}$

ii) $\{x : x \text{ is a positive integer less than } 12\}$

iii) $\{x : x \text{ is the square of an integer and } x < 100\}$

iv) $\{x : x \text{ is an integer such that } x^2 = 2\}$

v) $\{x : x \text{ is a non-negative integer less than } 5\}$

Write the set builder form for the following sets.

i) $\{0, 3, 6, 9, 12\}$

ii) $\{0, -3, -2, -1, 1, 2, 3\}$

$$iii) \{m, n, o, p\}$$

$$2) i) \{1, -1\}$$

$$ii) \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$$

$$iii) \{1, 4, 9, 16, 25, 36, 49, 64, 81\}$$

$$iv) \{\}$$

$$v) \{0, 1, 2, 3, 4\}$$

$$2) i) \{x : x \text{ is a whole number}\}$$

$$\{x / 0 \leq x \leq 10, x \text{ is a whole number and}$$

$$ii) \{x : x \text{ is an integer : } -3 \leq x \leq 3\}$$

$$iii) \{x : x \text{ is an alphabet less than 9}\}$$

10) Determine whether each of these pairs of sets are equal.

i) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}$

~~$\{5, 3, 1\}$~~

ii) $\{1, \{1\}\}$, $\{\{1\}\}$

⇒ i) members or elements are same. ∴ sets are same.

ii) they are not same cause $\{1\}$ is an element and a set of element.

11) Suppose $A = \{2, 4, 6\}$, $B = \{2, 6\}$
 $C = \{4, 6\}$ and $D = \{4, 6, 8\}$.

determine which of these sets are subsets of which other of these sets.

⇒ A is not subset of B cause 4 not present.

B is a subset of A.

C is a subset of D.

10/25
=

Relations

A relation is the subset of the Cartesian Product.

Representing Relations

Representing using matrices

A relation b/w finite sets

↓
A set with finite number of elements

Can be represented using a matrix.

Suppose R is a relation from $A = \{a_1, a_2, \dots, a_m\}$ to $B = \{b_1, b_2, \dots, b_n\}$.
 A has got m number of elements and B has got n number of elements. The Relation R can be represented by the matrix $M_R = \{m_{ij}\}$

where $m_{ij} = \begin{cases} 1, & (i, j) \in R \\ 0, & (i, j) \notin R \end{cases}$

1) Let $A = \{1, 2, 3\}$
 $B = \{1, 2\}$

Let R be the relation from A to B

Such that $R = \{(2,1), (3,1), (3,2)\}$

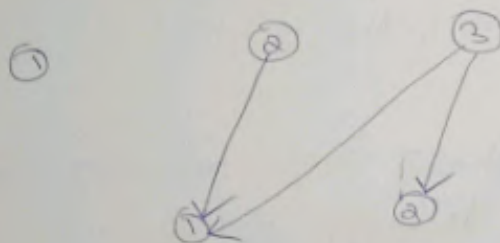
which is the matrix representation of R .

$\Rightarrow$

		1	2
1		0	0
2		1	0
3		1	1

$(2,1), (3,1), (3,2)$ is related

graphic representation.



a) Let R be a relation from set A to B . which is represented by the following matrix. identify the ordered pairs that are present in R .

			b_1	b_2	b_3	b_4	b_5
a_1		0	1	0	0	0	0
a_2		1	0	1	1	0	0
a_3		1	0	1	0	1	0

$$\text{let } A = \{a_1, a_2, a_3\}$$

$$B = \{b_1, b_2, b_3, b_4, b_5\}$$

$$R = \left\{ (a_1, b_2), (a_1, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_3), (a_3, b_1), (a_3, b_5), (a_3, b_4) \right\}$$

3) Find the matrix representing the following relation defined on set $\{1, 2, 3, 4\}$.

$$R = \{(1,1), (1,3), (2,1), (2,3), (2,4), (3,1), (3,2), (4,1)\}$$

4) Find the ordered pairs in the relation R represented by the following graphs.

